

Energy Shocks and the Green Adoption Channel in a Small Open Economy

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Abstract

European governments spent more than €600 billion shielding households from the 2021–23 energy shock, mostly by suppressing retail energy prices ([Bruegel, 2023](#)). We show that holding down the price also holds back the crisis-time adoption of green durables, a margin the macroeconomic literature has left out. We add a discrete brown/green durable choice to the small open economy HANK model of [Auclert et al. \(2024\)](#). An energy shock raises the relative price of brown energy and triggers an adoption wave: the green share rises by about eight percentage points at the peak, from a five-percent base. The switch is a lumpy purchase financed against a borrowing constraint, so adoption concentrates among unconstrained households. A retail price cap freezes the brown price and eliminates the wave; an income transfer of equal fiscal cost leaves the price free and preserves it, at comparable output stabilisation. The same suppression shuts down the expenditure-switching gain a small open economy draws from the shock. Two further results follow. An untargeted flat transfer dominates one indexed to pre-crisis energy use, because homothetic energy demand makes energy-indexed targeting regressive. A permanent carbon tax that greens the steady state provides no first-order crisis insurance. Monetary policy moves the transition only weakly, through the financing cost of the switch. Governments that shield households should support incomes and leave the retail price free.

Keywords: energy shocks; green transition; durable adoption; HANK; price caps; transfers; carbon taxation.

JEL codes: E21, E32, F41, H23, Q43, Q58.

1 Introduction

When wholesale energy prices tripled in 2021–22, European governments responded by shielding consumers. Between September 2021 and January 2023 EU member states committed more than €600 billion to protect households and firms, mostly through measures that hold down the retail price itself: caps, regulated tariffs, and energy tax cuts (Bruegel, 2023). These measures protected purchasing power, which was their purpose. But they act on the price that governs the energy transition. A retail price that is held fixed cannot signal scarcity, and households substitute away from fossil energy in response to that signal. Bruegel, which assembled the spending figures, drew the same conclusion and urged governments to replace price-suppressing measures, de facto fossil-fuel subsidies, with targeted income support (Bruegel, 2023).

This paper studies one margin of that trade-off which the macroeconomic literature has left out: the adoption of green durables during the crisis. A fossil price spike is the moment when switching to a technology that does not burn fossil energy pays best, and European households responded along this margin. Heat-pump sales reached a record of about three million units in 2022, up roughly 38% as gas prices surged. They fell 5% in 2023 and a further 21% in 2024 as gas prices normalised; the industry association named the low price of subsidised gas among the three main causes, and sales recovered only once governments cut electricity taxes (European Heat Pump Association, 2023, 2024, 2025). The margin responds to the relative price, and the price shields reached it.

We introduce this margin into an otherwise standard heterogeneous-agent model of an energy-importing economy. Households in the small open economy HANK of Auclert, Monneray, Rognlie, and Straub (2024; henceforth AMRS) choose discretely whether to pay a one-time resource cost for a green durable, a heat pump or an electric vehicle, that insulates them from the fossil price. The choice is a logit over the durable stock and is solved in the sequence space (Auclert et al., 2021). Everything else is inherited from Auclert et al. (2024): the CES energy nest, sticky import prices, the wage Phillips curve, and the constant-real-rate monetary baseline. The aggregate block is a generic energy importer; the extensive margin is the only new object, and we calibrate to the European episode (Section 4, Section 8). The closest antecedent is Dietrich et al. (2025), who shows in a representative-agent model that a carbon-pricing transition is mildly inflationary and that green durables are interest-rate sensitive. We keep that channel but add household heterogeneity, an energy shock, and the fiscal and open-economy blocks. Our question is not how a transition unfolds but how a crisis, and the policies that meet it, move the adoption margin.

The central result is a non-neutrality. An energy shock endogenously triggers an adoption

wave: in the baseline crisis the green share rises by about eight percentage points at the peak, from a five-percent base. The two ex-post instruments in the recent literature, a retail price cap and a Slutsky-compensating transfer indexed to pre-crisis energy use, stabilise output comparably but differ on the new margin. The cap holds the brown retail price at its pre-crisis level and thereby switches adoption off: the peak green-share response falls from eight points to essentially zero. The transfer supports incomes and leaves the wave intact. Shielding consumers from the relative price is therefore not neutral for the transition, and the cost is the loss of the extensive margin itself rather than a marginal distortion of intensive-margin demand. The mechanism is the brown retail price: the cap fixes it and the transfer leaves it free, so the ranking survives every other modelling choice (Section 6).

The second question is who should be shielded, and how. Energy demand is close to proportional to total consumption within technology groups. A transfer indexed to pre-crisis energy use therefore hands the largest cheques to the wealthiest households, which have the lowest MPCs and gain the least insurance per euro. An untargeted flat transfer of the same fiscal cost reaches the high-MPC households the crisis hits hardest and delivers strictly higher welfare (a consumption-equivalent variation of -0.36% against -0.84%), while leaving adoption intact. This result is conditional on homothetic energy demand. With a declining Engel curve for energy, as in [Bayer et al. \(2026\)](#) through energy-share heterogeneity or in [Langot et al. \(2026\)](#) through a subsistence floor, low-income households carry a larger energy share and the ranking could weaken. We take the homothetic nest from [Auclert et al. \(2024\)](#) and leave that case aside. Under constant energy shares, energy-indexed targeting is dominated by an equal-cost flat transfer.

The third question moves from ex-post to ex-ante policy. Europe will soon run the ex-ante experiment deliberately: the forthcoming ETS2 carbon price for buildings and road transport will raise the household brown-energy price by design ([European Commission, 2026](#)). A permanent carbon tax that greens the steady state before the crisis is a poor substitute for ex-post insurance. Its crisis dividend is zero to first order, and it does not improve as shocks become more persistent. The prepared economy in fact adopts *more* during the crisis; what falls is the welfare value of the marginal switch, because pre-greening uses up, before the shock, an adjustment margin that is most valuable when the economy enters the crisis with room to green (Section 7). The externality and the standing gain of a greener steady state justify the carbon tax; insurance against the next shock does not.

Two auxiliary results are conditional, and we state the conditions rather than resolve them. The sign of the cap’s private-welfare effect depends on the energy-supply elasticity, which locates one axis of the disagreement between [Langot et al. \(2026\)](#) and [Bayer et al. \(2026\)](#) without nesting their models. The sign of the adoption channel’s contribution to

output depends on how the green resource flows are booked in the balance of payments; we report the import booking throughout and collect the alternative in Appendix A. The policy ordering is robust to both choices, since it follows from the relative price. The size of the adoption wave, by contrast, depends on one parameter, the logit taste scale σ_ε , and we discipline it with data rather than assume it. At a five-percent green share every brown household sits in the exponential tail of the logit, so the semi-elasticity of switching is $1/\sigma_\varepsilon$ household by household (Appendix D). Matching the response of adoption to a \$1,000 incentive in the quasi-experimental vehicle literature (Gallagher and Muehlegger, 2011; Chandra et al., 2010; Muehlegger and Rapson, 2022) pins $\sigma_\varepsilon \in [0.05, 0.07]$, with an implied green premium of \$15,000–\$18,000. The steady state and the policy ordering do not depend on the scale; the size of the wave does.

Section 2 places the paper in the literature. Section 3 presents the model and Section 4 the calibration, the identification of the taste scale, and the experimental design (the two energy shocks, the frozen-choice counterfactual that isolates the adoption channel, the policy instruments, and the welfare measure). Section 5 documents the crisis adoption wave and its contribution to the aggregate response. Section 6 compares the ex-post instruments, prices against incomes, and Section 7 the ex-ante carbon tax. Section 8 ranks the instruments by welfare and states what the results depend on. Section 8 maps the results into European policy, and Section 9 concludes.

2 Related literature

We draw on three bodies of work: energy-shock stabilisation in heterogeneous-agent open economies; the macroeconomics of climate policy and household technology adoption; and the microeconometrics of durable adoption that disciplines the margin we add.

2.1 Energy-shock stabilisation in heterogeneous-agent economies

The closest work studies fiscal and monetary responses to the 2021–23 energy shock in open HANK economies. Auclert et al. (2024) provide the aggregate frame we inherit, an energy-importing HANK with a CES energy nest, sticky import prices, and intertemporal energy suppliers. In their setting a low substitution elasticity makes the shock contractionary through the real-income channel, so a subsidy stabilises output, inflation and inequality at once; the drawback is a cross-country externality, since coordinated subsidies spike the world price. Bayer et al. (2026) build a two-country monetary union to study that spillover, with a richer medium-scale block, two assets, capital, energy in production, and a distributional

mechanism that runs through within-country energy-share heterogeneity: poorer households spend more on energy and suffer larger real-income losses. [Langot et al. \(2026\)](#) take France as a small economy inside the euro area, fed by the government’s official energy-price forecasts, with a higher intensive substitution elasticity and a subsistence energy floor that makes demand non-homothetic. On a conditional-forecast path they find the French shield effective, largely through a labour-cost channel that contains the price–wage spiral.

These models disagree on the sign of the shield’s welfare effect, and the supply closure organises the disagreement. Under perfectly elastic supply ([Langot et al., 2026](#)) the shield is effective; under a fixed union-wide energy quantity ([Bayer et al., 2026](#)) price suppression is destructive, because it blocks the substitution scarcity requires. Our two supply closures, elastic price-taking and fixed-quantity, reproduce the sign each obtains, which locates one source of the contrast in a single parameter, the supply elasticity at the relevant horizon. The parallel has limits. With no second country we cannot represent Bayer’s cross-border transmission, and our wage block, lacking an investment or employment margin, reproduces the sign of Langot’s pro-shield result but not its full labour-cost mechanism.

None of these models contains an endogenous energy technology. AMRS, Bayer, and Langot hold the household’s energy type fixed, so the shield’s cost to the transition is invisible to them. The nearest antecedent on the technology margin is [Audzei and Sutóris \(2026\)](#), who add continuous energy-conservation capital to a stylised heterogeneous-agent New Keynesian economy and show that the monetary stance shapes households’ incentive to build resilience against energy price shocks. Their margin is an *intensity* of energy use: there is no fossil/green technology type, hence no relative energy price for a cap to suppress and no composition of the durable stock for fiscal policy to move. Our contribution is to make that type a choice. The adoption margin adds a cost of price suppression the aggregate literature cannot see: capping the brown price switches the transition off. The AMRS real-income channel still operates in the background of every experiment. Our heterogeneity is over discount factors and the adoption margin rather than over energy shares, so we are silent on the energy-share-of-the-poor mechanism; the two distributional channels complement each other.

2.2 Climate policy and household technology adoption

The second literature studies the macroeconomics of climate policy, and in particular its interaction with monetary policy. Governments have pursued opposite strategies to move households onto clean durables: the United States leans on subsidies and tax credits, the European Union increasingly on carbon pricing. The instrument closest to our setting is

the forthcoming carbon price for buildings and road transport (ETS2), which raises the retail price of household fossil energy by design (European Commission, 2026). Our crisis raises the same brown price, and our ex-ante experiment moves it (Section 7). Dietrich et al. (2025) are the closest antecedent on the monetary side: in a representative-agent model with brown and green durables, a carbon-pricing transition is mildly inflationary and green-durable purchases are markedly interest-rate sensitive, so a strict inflation target slows decarbonisation. We keep that price-stability channel and add what a representative agent cannot carry: household heterogeneity, an energy shock, and the fiscal and open-economy blocks that let us rank crisis instruments.

A distributional strand asks who bears the cost of climate policy and how revenue recycling reshapes it (Fried et al., 2018, 2024; Ascari et al., 2025); Klenert et al. (2018) review the recycling design that makes carbon pricing politically durable. Closest to us are the papers that let households choose the technology. Kuhn and Schlattmann (2024) study green subsidies when households replace a polluting old durable with a clean new one and find that subsidies should target high-income households. Crucitti et al. (2024) place heat-pump adoption in a life-cycle model with liquidity constraints and show that financial frictions are the binding barrier, so installation subsidies favour poorer households while electricity-tax cuts favour richer ones. Fried et al. (2025) build emissions from home-production tasks with clean and dirty equipment and heterogeneous distaste for emissions, and find that subsidies should target equipment at intermediate adoption and be withdrawn from the highest incomes. On the firm side, Lanteri and Rampini (2025) model extensive-margin investment in cleaner capital under credit constraints.

These papers adopt a long-run, transition perspective and study adoption in the absence of a macroeconomic disturbance; with the exception of Fried et al. (2025) they abstract from general equilibrium, and none carries monetary policy. We embed the household adoption margin in a full HANK economy hit by an energy shock, and study how carbon pricing, green subsidies, transfers and the monetary stance jointly move the pace of adoption and the transmission of the shock. The question shifts from how a policy-driven transition unfolds in calm times to how a crisis, and the instruments that meet it, trade off softening the short-run cost of the shock against preserving the adjustment that reduces future exposure to it.

2.3 Durable adoption and its identification

The adoption margin is a discrete durable choice inside an incomplete-markets household. We solve it with the discrete-continuous endogenous grid method of Iskhakov et al. (2017),

whose extreme-value taste shocks smooth the choice into differentiable Jacobians for the sequence-space solution of [Auclert et al. \(2021\)](#). In our calibration the taste shocks do double duty: at a rare adoption margin the logit scale is the structural semi-elasticity of switching (Appendix D), so it must be identified rather than set small.

We discipline it against the quasi-experimental literature on durable adoption. [Gallagher and Muehlegger \(2011\)](#) and [Chandra et al. \(2010\)](#) estimate that a \$1,000 tax incentive raises hybrid-vehicle adoption by 31–38% among higher-income early adopters; [Muehlegger and Rapson \(2022\)](#) estimate a demand elasticity of -2.1 for electric vehicles among low- and middle-income households, roughly 8% per \$1,000 at their mean transaction price, the mass-market floor on responsiveness. [Beresteanu and Li \(2011\)](#) document the operating-cost margin, the response of hybrid demand to gasoline prices, which is the analogue of our energy-cost channel. These are estimates of the upfront- and operating-cost elasticities the model needs, and Section 4 maps them onto the taste scale. The structural discrete-choice tradition that estimates such scales from rare, lumpy adjustment ([Rust, 1987](#)) is the right reference point for our regime. At interior-probability margins such as labour-force participation, the smoothing reading applies and the scale is a numerical convenience.

3 The model

The economy is a small open economy that imports its fossil energy, populated by heterogeneous households, competitive firms producing a domestic good from labour, a union that sets wages, a government, and a rest of the world that sets world prices. With one exception, the environment is the small open economy HANK of [Auclert et al. \(2024\)](#).

The base is the [Auclert et al. \(2024\)](#) model. We change one thing, in the household block: the price of energy a household faces is no longer given but depends on a durable it chooses. Each household holds a brown or a green durable (a gas boiler or a heat pump, a combustion or an electric vehicle) and pays a one-time cost ψ_g to switch from brown to green, after which it buys energy at the lower green price. The change is confined to the household’s problem, but it reaches the rest of the model through the price it controls: the energy price sits inside the consumption bundle and the budget constraint. Adoption is an intertemporal choice: the household weighs the upfront cost ψ_g against the stream of energy savings the green durable delivers over its life, and pays ψ_g out of current resources against the borrowing constraint; only households with enough liquidity can afford it. A shock or a policy that moves the brown energy price moves this decision, and with it the whole household block. Everything outside that, the CES nest, sticky import prices, the wage Phillips curve, the mutual fund, and the constant-real-rate rule, follows AMRS.

Section 3.1 presents the household block. Sections 3.2–3.5 lay out the production side, the price system, the open-economy accounts, and the government, which are standard. Section 3.6 defines equilibrium, and Section 3.7 the steady state; the two crisis experiments are defined in Section 4.2. Time is discrete and quarterly. A bar denotes a steady-state value; a date subscript indexes the period, with x_{t-1} a lag and x_{t+1} an expected lead. For the durable, d_{t-1} is the technology chosen last period, d_t^- the technology held at the start of t after breakdown, and d_t the technology chosen. All retail prices are expressed relative to the CPI P and carry the subscript $/P$.

3.1 Households

There are three permanent discount-factor types $i \in \{0, 1, 2\}$ of equal mass, with discount factors β_i chosen in Section 4 to match aggregate MPCs. A household is described by three state variables: its assets a , its idiosyncratic labour productivity e , and the durable technology $d \in \{B, G\}$ it operates. Productivity follows a discretised log-AR(1), $\log e_{t+1} = \rho_e \log e_t + \epsilon_{t+1}$, with a mean-one grid and transition matrix Π , as in Auclert et al. (2024). The technology is the new object, and because the paper concerns a decision over it, we fix when, within a period, the household holds it, loses it, and chooses it.

3.1.1 The technology and the timing of the choice

A household enters the period operating the technology it chose last period, which we denote d_{t-1} . Three events then occur, in this order, before it consumes.

1. *Breakdown.* A green durable fails with probability δ_g and reverts to brown; a brown durable is unaffected. We write d_t^- for the technology the household holds *after* this draw: the technology it brings to the decision. Hence $d_t^- = B$ whenever $d_{t-1} = B$, while $d_t^- = G$ with probability $1 - \delta_g$ when $d_{t-1} = G$.
2. *Productivity.* The household draws its labour productivity e_t .
3. *The decision.* The household chooses, in a single problem, the technology d_t it will operate together with its consumption and saving. A household holding brown ($d_t^- = B$) may keep brown or pay the switching cost ψ_g to adopt green; a household holding green ($d_t^- = G$) operates it, since running an owned green durable costs nothing and reverting to brown is never chosen.

No information arrives between the productivity draw and the decision, so the household settles technology, consumption, and saving *jointly* (Section 3.1.3). The technology it oper-

ates is what it carries forward, so d_t becomes the technology held on entry next period; the within-period sequence is

$$\underbrace{d_{t-1}}_{\text{held on entry}} \xrightarrow{\text{breakdown}} \underbrace{d_t^-}_{\text{held at the decision}} \xrightarrow{\text{choice}} \underbrace{d_t}_{\text{operated, carried forward}} .$$

Two binary objects thus describe the technology: the holding at the decision, d_t^- , and the operated technology, d_t . The switching cost is paid exactly when a brown holder adopts, that is when $d_t^- = B$ and $d_t = G$; we write $\mathbb{I}[d_t^- = B, d_t = G]$ for a switcher and $\mathbb{I}[d_t = G]$ for a household operating green. The feasible choices are

$$\mathcal{F}(d_t^-) = \begin{cases} \{B, G\} & d_t^- = B \quad (\text{keep brown, or pay } \psi_g \text{ to adopt}), \\ \{G\} & d_t^- = G \quad (\text{operate the green durable}). \end{cases}$$

Two features follow directly. First, the technology is persistent but asymmetric: brown is absorbing, and green is lost only through breakdown. Second, the adoption decision is an option held by the brown population alone: a green household has nothing to choose.

A constant green share therefore requires a standing flow of switchers to replace the durables that fail. In steady state this flow is

$$\bar{D}^{\text{sw}} = \delta_g \bar{D}^G, \tag{1}$$

with D^G and D^{sw} the population shares of green households and of switchers. At the calibrated $\bar{D}^G = 5\%$ and a five-year durable life, this flow is about a quarter of one percent of households per quarter: adoption is a rare, lumpy event, so the object to identify is its elasticity rather than its level (Section 4.1, Appendix D).

3.1.2 Preferences and the cost of living

Labour supply is determined at the union level (Section 3.2), so flow utility is CRRA over a consumption bundle c alone,

$$u(c) = \frac{c^{1-\sigma}}{1-\sigma}, \quad u(c) = \log c \quad \text{for } \sigma = 1, \tag{2}$$

with σ the inverse elasticity of intertemporal substitution, as in Auclert et al. (2024). The bundle is the Auclert et al. (2024) CES nest of energy against a home/foreign goods composite, with energy weight α_E and a low substitution elasticity η_E .

The departure from AMRS is that energy comes at a technology-specific price: a house-

hold operating brown buys energy at P_B^E , one operating green at $P_G^E < P_B^E$. The operated technology therefore fixes both the energy price and the exact cost-of-living index,

$$p_{d,t}^E = P_{B,t}^E + \mathbb{I}[d = G] (P_{G,t}^E - P_{B,t}^E), \quad (3)$$

$$\frac{p_{G,t}}{p_{B,t}} = \left[\frac{(1 - \alpha_E) p_{HF/P,t}^{1-\eta_E} + \alpha_E (P_{G,t}^E)^{1-\eta_E}}{(1 - \alpha_E) p_{HF/P,t}^{1-\eta_E} + \alpha_E (P_{B,t}^E)^{1-\eta_E}} \right]^{\frac{1}{1-\eta_E}}, \quad \frac{p_{G,t}}{p_{B,t}} = (P_{G,t}^E / P_{B,t}^E)^{\alpha_E} \text{ for } \eta_E = 1. \quad (4)$$

The adoption choice responds to the relative saving $p_G/p_B < 1$, the cheaper cost of living the switching cost buys; at the calibrated steady state $p_G/p_B \approx 0.981$.

3.1.3 The household's problem

A household of type i reaches period t with entry assets a_{t-1} and productivity e_t , holding the post-breakdown durable d_t^- , and chooses its technology d_t , consumption c_t , and saving a_t . Its resources are three. Assets earn the ex-post real return r_t , the return on the mutual fund of Section 3.2. This return equals the lagged ex-ante rate r_{t-1}^{ante} at every date but the first; at date 0 the revaluation of the fund's equity drives the two apart (equation (13)), and the household bears that surprise. Labour income is $e_t Z_t$, where $Z_t \equiv (1 - \tau_t^L) w_t n_t$ is aggregate after-tax labour income distributed in proportion to productivity, as in [Auclert et al. \(2024\)](#); the household does not choose hours, which the union sets (Section 3.2). The government may pay a transfer T_t , specified below. On the spending side, the household buys the consumption bundle at its technology-specific cost of living $p_{d,t}$, saves $a_t \geq 0$ subject to the borrowing constraint, and, if it is a brown holder adopting green this period, pays the switching cost ψ_g , once. The budget constraint collects these items,

$$p_{d,t} c_t + a_t + \psi_g \mathbb{I}[d_t^- = B, d_t = G] = (1 + r_t) a_{t-1} + e_t Z_t + T_t. \quad (5)$$

The switching cost is paid exactly once, in the period a brown holder adopts, and it buys the permanently cheaper bundle $p_G < 1$ for as long as the durable survives: an upfront, lumpy outlay against a gradual operating saving. Because the outlay is financed out of current resources against the borrowing constraint, wealth governs who can undertake it, which is why adoption concentrates among the unconstrained (Section 5.1).

The transfer carries ex-post fiscal policy,

$$T_t = T_t^{\text{unt}} + \text{ins}_E (p_t^E - \bar{p}^E) \bar{c}_{E,i}^B(e, a), \quad (6)$$

an untargeted lump sum T^{unt} plus a targeted (Slutsky) transfer that scales, with insurance

proportion ins_E , the crisis rise in the energy price by the household's *pre-crisis* brown-energy demand $\bar{c}_{E,i}^B(e, a)$; both are zero in the baseline. One design feature matters throughout: the transfer conditions on steady-state energy use, never on the current technology. A household that switches forfeits nothing, so the transfer leaves the adoption margin undistorted, in contrast to the price cap of Section 3.3, which acts on P_B^E itself.

The household chooses its technology, consumption, and saving *together*, in a single problem:

$$\max_{d_t \in \mathcal{F}(d_t^-)} \max_{c_t, a_t \geq 0} \left\{ u(c_t) + \beta_i \mathbb{E}_t [\bar{V}_{t+1}(d_{t+1}^-, e_{t+1}, a_t)] \right\} \quad \text{subject to (5),} \quad (7)$$

with $\bar{V}_{t+1}$ the continuation value (defined in (10)) and next period's holding d_{t+1}^- given by breakdown applied to the operated technology d_t . The nested maxima do not describe a temporal sequence, since nothing is realised between them; they are the structure of a discrete–continuous choice: for each feasible technology the household solves the continuous consumption-saving problem, obtaining the technology-conditional value $V_t(d_t \mid d_t^-, e_t, a_{t-1})$, in which the origin d_t^- enters only through the switching cost; it then selects the technology (Section 3.1.4). Only three continuous problems are ever relevant: a brown household (p_B , no switching cost), a green incumbent (p_G , no switching cost), and a green adopter (p_G , resources reduced by ψ_g). Each satisfies the usual Euler and envelope conditions,

$$u'(c_t) = \beta_i p_{d,t} \mathbb{E}_t [\bar{V}_{a,t+1}(d_{t+1}^-, e_{t+1}, a_t)], \quad V_{a,t} = (1 + r_t) u'(c_t) / p_{d,t}, \quad (8)$$

with $u'(c_t) = c_t^{-\sigma}$; a green household's cheaper bundle raises the consumption value of every unit of wealth. CES demands split by the operated technology,

$$c_{E,t}(d) = \alpha_E (p_{d,t}^E / p_{d,t})^{-\eta_E} c_t, \quad c_{HF,t}(d) = (1 - \alpha_E) (p_{HF/P,t} / p_{d,t})^{-\eta_E} c_t, \quad (9)$$

and brown versus green energy use follow the technology, $c_{E,t}^B = c_{E,t}(d_t) (1 - \mathbb{I}[d_t = G])$ and $c_{E,t}^G = c_{E,t}(d_t) \mathbb{I}[d_t = G]$.

3.1.4 The technology choice

The outer maximum in (7) selects the technology, completing the joint problem. A household holding d_t^- compares the technology-conditional values $V_t(d_t \mid d_t^-, \cdot)$; with an i.i.d. extreme-value taste shock of scale σ_ε on each option, the choice is logit and the household's

continuation value $\bar{V}_t$ is the log-sum-exp,

$$\begin{aligned}\Pr_t(d_t \mid d_t^-, e_t, a_{t-1}) &= \frac{\exp\{V_t(d_t \mid d_t^-, e_t, a_{t-1})/\sigma_\varepsilon\}}{\sum_{d' \in \mathcal{F}(d_t^-)} \exp\{V_t(d' \mid d_t^-, e_t, a_{t-1})/\sigma_\varepsilon\}}, \\ \bar{V}_t(d_t^-, e_t, a_{t-1}) &= \sigma_\varepsilon \log \sum_{d' \in \mathcal{F}(d_t^-)} \exp(V_t(d' \mid d_t^-, e_t, a_{t-1})/\sigma_\varepsilon).\end{aligned}\tag{10}$$

For a green holder the feasible set is a singleton, so $\bar{V}_t(G, \cdot) = V_t(G \mid G, \cdot)$ and there is nothing to decide. The only genuine choice is the brown holder's: a two-way comparison of staying brown against paying ψ_g to adopt green. That cost already sits inside the adopter's budget constraint (5), so the option values price it automatically.

The economics of the margin is simplest in steady state. The marginal switcher equates the premium to the discounted operating-cost saving over the durable's expected life,

$$\psi_g \approx \frac{(\bar{P}_B^E - \bar{P}_G^E) \bar{C}_E^G}{r + \delta_g},\tag{11}$$

which ties the adoption threshold to the delivered cost ratio $\varrho = \bar{P}_G^E / \bar{P}_B^E$. Every experiment in the paper acts on this one comparison. An energy crisis raises P_B^E and lifts the return to switching; a retail price cap holds P_B^E at its pre-crisis level and removes that return entirely; a transfer supports income while leaving P_B^E , and hence (11), free; a permanent carbon tax raises the brown price in steady state, greening the economy before any crisis arrives.

Away from steady state, the responsiveness of the margin is governed by σ_ε . The taste shocks are, in the tradition of [Iskhakov et al. \(2017\)](#), the smoothing that makes the discrete choice differentiable; at a rare margin they are also the structural semi-elasticity of switching, so σ_ε cannot be treated as a nuisance parameter and set small. Section 4.1 identifies it from quasi-experimental adoption elasticities, and Appendix D develops the smoothing-elasticity duality.

3.1.5 Aggregation

For each type i , aggregates integrate the household policies over the stationary distribution Λ_i , and the economy averages the three types equally,

$$X_t = \frac{1}{3} \sum_i X_{i,t}, \quad X \in \{C, A, D^G, D^{\text{sw}}, C_E^B, C_E^G, C_{HF}, \dots\}.\tag{12}$$

Aggregate energy demand is $c_{E,t} = C_{E,t}^B + C_{E,t}^G$, and home- and foreign-good demands apply the common inner nest to the composite, $c_{H,t} = (1 - \alpha_F) p_{H/HF,t}^{-\eta} C_{HF,t}$ and $c_{F,t} =$

$\alpha_F p_{F/HF,t}^{-\eta} C_{HF,t}$. The two aggregates the policy experiments track are the green share D^G , the population share operating green, and the switching flow D^{sw} , the share adopting green each period.

3.2 Production and wage setting

Production and profits. Competitive firms produce the domestic final good from labour with productivity A^N , sold at a constant gross markup μ , so output is $y = A^N n$ and after-tax aggregate labour income is $Z = (1 - \tau^L) w n$, with n hours and w the real wage. Monopoly profits are capitalised into a mutual fund that households hold,

$$D_t = (\mu - 1) w_t n_t, \quad J_t = D_t + \frac{J_{t+1}}{1 + r_t^{\text{ante}}}, \quad j_t = \frac{J_{t+1}}{1 + r_t^{\text{ante}}}, \quad 1 + r_t = \frac{j_t}{j_{t-1}}, \quad (13)$$

following the notation of [Auclert et al. \(2024\)](#): r_t^{ante} is the ex-ante real rate at which the risk-neutral fund prices its claims (equation (27)), j is the ex-dividend share price, and r is the fund's ex-post return: the return households earn in (5). Because the fund equates ex-ante returns across assets, the ex-post return equals the lagged ex-ante rate at every date but the first, $r_t = r_{t-1}^{\text{ante}}$ for $t \geq 1$; only the impact return r_0 differs, because the date-0 revaluation of the fund's equity feeds the revaluation term in (26). Importers are foreign-owned and the energy endowment is not held domestically, so domestic firms are the only domestically held risky claim.

Wage setting. A union sets the nominal wage under Calvo frictions θ_w , trading off households' marginal rate of substitution against the after-tax wage, with a real-wage-stabilisation term (weight ζ_{BG}) as in [Auclert et al. \(2024\)](#). With slope $\kappa_w = (1 - \theta_w)(1 - \beta\theta_w)/\theta_w$, inverse Frisch elasticity φ , and labour-disutility constant v_φ , wage inflation follows

$$\pi_t^w (1 + \pi_t^w) = \beta \pi_{t+1}^w (1 + \pi_{t+1}^w) + \frac{\kappa_w}{v_\varphi n_t^\varphi} \left[v_\varphi n_t^\varphi - \frac{(1 - \tau_t^L) w_t C_t^{-\sigma}}{\mu} - \zeta_{BG} \frac{(w_t - \bar{w}) w_t \bar{C}^{-\sigma} \bar{n}}{\mu n_t \bar{w}} \right], \quad (14)$$

and the real wage evolves as $w_t = (1 + \pi_t^w) w_{t-1} / (1 + \pi_t)$.

3.3 The price system

The paper's mechanism runs entirely through relative prices, so we lay the price system out in one place. Table 1 collects it: world prices at the top, a wholesale layer carrying the nominal rigidities, a retail layer where policy acts, and the CPI that aggregates them.

Table 1: The price system

Symbol	Object	Determined by
P^{E*}, P_F^*, C^*, r^*	world energy and goods prices, demand, rate	exogenous ^a
p^E	wholesale brown energy price	NKPC (17)
P_B^E	brown households' energy price (tax-inclusive)	cap rule and τ_b , eq. (18)
P_G^E	green households' energy price (tax-inclusive)	world price and τ_g , eq. (19)
$P_B^{E,\text{pre}}, P_G^{E,\text{pre}}$	pre-tax energy prices (BoP, carbon account)	eqs. (18)–(19)
$p_{H/P}$	domestic good	markup pricing (15)
$p_{F/P}$	imported consumption good	NKPC (16)
$p_{HF/P}, p_{H/HF}$	goods composites	outer nest (21)
p_B, p_G	brown/green cost of living	bundle nests (20); saving p_G/p_B (4)
Q	real exchange rate	UIP (22)

Notes: All retail prices are relative to the CPI. ^aUnder the fixed-quantity closure P^{E*} is the price that clears the fixed regional quantity (Section 3.6).

The domestic good. Marginal-cost pricing at the constant markup gives

$$p_{H/P,t} = \frac{\mu}{A^N} w_t. \quad (15)$$

Imported goods and wholesale energy. Retailers of the imported good and of energy reset prices under Calvo frictions θ_F and θ_E , so world price movements pass through to domestic retail prices only gradually. With the usual slopes κ_F and κ_E ,

$$\kappa_F \left(\frac{Q_t P_F^*}{p_{F/P,t}} - 1 \right) + \beta_F \pi_{t+1}^F (1 + \pi_{t+1}^F) - \pi_t^F (1 + \pi_t^F) = 0, \quad (16)$$

$$\kappa_E \left(\frac{Q_t P_t^{E*}}{p_t^E} - 1 \right) + \beta_E \pi_{t+1}^E (1 + \pi_{t+1}^E) - \pi_t^E (1 + \pi_t^E) = 0. \quad (17)$$

The margins these retailers earn out of steady state accrue abroad (the importers are foreign-owned, as in Auclert et al., 2024); they matter below only because the import bill is valued at retail rather than world prices.

Retail energy prices. Each technology's energy price is built in two steps: a pre-tax price, which values the resource flows in the balance of payments and the carbon account, and the tax-inclusive price the household pays. For brown energy the pre-tax price applies

the cap to the wholesale price, and the carbon tax scales it,

$$P_{B,t}^{E,\text{pre}} = (1 - \tau^E) p_t^E + \tau^E \bar{p}^E, \quad P_{B,t}^E = P_{B,t}^{E,\text{pre}} (1 + \tau_b), \quad (18)$$

so τ^E governs pass-through continuously: at $\tau^E = 0$ households face the wholesale price, at $\tau^E = 1$ the pre-tax brown price is frozen at its pre-crisis level: the full shield. Green energy is imported at the constant world price $P_G^{E*} = \varrho \bar{p}^E$, $\varrho < 1$, and carries its own carbon tax τ_g ,

$$P_{G,t}^{E,\text{pre}} = Q_t P_G^{E*}, \quad P_{G,t}^E = P_{G,t}^{E,\text{pre}} (1 + \tau_g), \quad (19)$$

so adopters are fully insulated from the fossil shock: the real exchange rate does not move under either shock, and P_G^E stays at $\varrho \bar{p}^E$.

The household side of the model reads only the tax-inclusive prices: the bundle nests (20), the budget (5) and hence the adoption margin (11) all price energy at P_B^E and P_G^E . The pre-tax prices reappear where resources cross the border, in the balance of payments (Section 3.4), and in the carbon account of Section 3.5. In the baseline $\tau_b = \tau_g = 0$ and the two coincide; under the carbon-tax experiments of Section 7 they diverge, and the distinction is what gives the tax real incidence: it raises the price the adoption margin compares without changing the resource cost of energy to the country.

The consumer price index. Each technology has its own consumption bundle. Writing p_B, p_G for the brown and green bundle prices relative to the CPI, the inner (energy versus goods) CES nests are

$$p_{d,t}^{1-\eta_E} = (1 - \alpha_E) p_{HF/P,t}^{1-\eta_E} + \alpha_E (P_{d,t}^E)^{1-\eta_E}, \quad d \in \{B, G\}, \quad (20)$$

with the outer (home versus foreign) nest

$$(1 - \alpha_F) p_{H/HF,t}^{1-\eta} + \alpha_F p_{F/HF,t}^{1-\eta} = 1. \quad (21)$$

The CPI is the expenditure-weighted index of the two bundles: P is defined by the requirement that nominal consumption expenditure deflated by P equal real consumption, $\int p_{d(i)} c(i) di = \int c(i) di$, so that $\sum_d \omega_{d,t} p_{d,t} = 1$ with $\omega_{d,t}$ the consumption share of state d . As green diffuses the energy leg is reweighted from brown toward green through $\omega_{G,t} = D_t^G$, so P is the true cost-of-living index the monetary rule reads, with no measurement gap between the price index and the basket households actually buy.

3.4 The open economy

Foreign demand and the exchange rate. Foreign demand for the home good is $c_H^* = \alpha^*(p_H^*)^{-\gamma} C^*$, with trade elasticity γ , and the real exchange rate is pinned by uncovered interest parity,

$$\frac{Q_t}{Q_{t+1}}(1 + r_t^{\text{ante}}) = 1 + r^*. \quad (22)$$

World energy supply. World supply is a fixed endowment E^* plus intertemporal stock arbitrage with cost Γ ,

$$E_t^{\text{stock}} = \frac{P_{t+1}^{E^*}/(1 + r^*) - P_t^{E^*}}{\Gamma}, \quad E_t^s = E_t^* + (E_{t-1}^{\text{stock}} - E_t^{\text{stock}}). \quad (23)$$

The home economy owns none of the endowment, being a pure importer, so energy rents never enter domestic wealth.

The balance of payments and the adoption flows. The adoption margin adds exactly two terms to the current account, and they pull in opposite directions. The switching cost is a resource outflow, booked as an import M^{sw} : it is front-loaded, paid once per switcher, and scales with the switching volume. The adopters' saved fossil energy is a resource inflow, booked as an import saving S^E : it accrues gradually, over the life of each green durable,

$$M_t^{\text{sw}} = \psi_g D_t^{\text{sw}}, \quad S_t^E = (P_B^{E,\text{pre}} - P_G^{E,\text{pre}}) C_{E,t}^G. \quad (24)$$

The energy saving is valued at pre-tax prices: the carbon taxes are transfers between domestic households and the government, not resource flows to the rest of the world. Their revenue, $R^{\text{carbon}} = \tau_b P_B^{E,\text{pre}} C_E^B + \tau_g P_G^{E,\text{pre}} C_E^G$, accrues to the government budget (Section 3.5) and never enters net exports. Net exports collect trade in goods and energy together with the two adoption flows,

$$nx_t = Q_t p_H^* c_{H,t}^* - p_{F/P,t} c_{F,t} - p_t^E c_{E,t} - M_t^{\text{sw}} + S_t^E, \quad (25)$$

and net foreign assets accumulate at the ex-ante rate, with one additional term,

$$nfa_t = nx_t + (r_t - r_{t-1}^{\text{ante}})(A_{t-1} - j_{t-1}) + (1 + r_{t-1}^{\text{ante}}) nfa_{t-1}. \quad (26)$$

The middle term is a revaluation. Households earn the ex-post return r on their wealth, while the country's accounts accrue at the promised ex-ante rate r_{t-1}^{ante} ; the two differ only at date 0, when the shock revalues the equity of the mutual fund (equation (13)). Applying the

surprise $r_t - r_{t-1}^{\text{ante}}$ to households' wealth net of the domestically held fund, $A_{t-1} - j_{t-1}$, records the one-off capital gain or loss on the country's external claims. At date 0, the only date the term is active, $A_{t-1} - j_{t-1}$ equals the net foreign position itself, since government debt is zero at the pre-crisis steady state; from date 1 onward $r_t = r_{t-1}^{\text{ante}}$ and the term vanishes.

Booking the adoption outlay as an import is a convention rather than a result. The baseline sets the import share of the switching bundle to one, $\alpha_F^{\text{switch}} = 1$, so the full cost $\psi_g D^{\text{sw}}$ enters (25) as M^{sw} ; Appendix A traces the continuous import share from one down to zero, where the outlay is spent entirely on the home good. The convention fixes the sign of the adoption channel's *output* contribution but none of the policy rankings, which run through relative prices alone.

3.5 The government

Monetary policy. The nominal rate follows a rule in current and expected inflation, with the real rate defined by the Fisher equation,

$$1 + i_t = (1 + \bar{r}^*)(1 + \phi_\pi \pi_t)(1 + \phi_{\pi e} \mathbb{E}_t \pi_{t+1}) + \varepsilon_t^m, \quad r_t^{\text{ante}} = \frac{1 + i_t}{1 + \mathbb{E}_t \pi_{t+1}} - 1, \quad (27)$$

where ε^m is a monetary shock, zero outside the monetary experiment of Appendix C. The baseline is the constant-real-rate rule of Auclert et al. (2024), $(\phi_\pi, \phi_{\pi e}) = (0, 1)$. Through (22) it pins the path of the real exchange rate, so a domestic transfer has no terms-of-trade effect, a property inherited from AMRS rather than assumed anew. A standard Taylor rule, $(\phi_\pi, \phi_{\pi e}) = (1.5, 0)$, is reported as a robustness contrast in Appendix C.

Fiscal policy. Fiscal policy is the object of Sections 6–7. Four instruments are deployed once the shock hits and are zero in the baseline; a fifth, the carbon tax, is permanent and defines the “prepared” economy of Section 7. Table 2 states, for each, what it acts on, the base its cost scales with, and its budgetary cost.

Table 2: Fiscal instruments

Instrument	Acts on	Cost scales with	Fiscal cost
Price cap (τ^E)	brown retail price P_B^E	actual energy use	$\tau^E(p^E - \bar{p}^E) c_E$
Slutsky transfer (ins_E)	household income	pre-crisis brown use	$ins_E(p^E - \bar{p}^E) \bar{C}_E^B$
Flat transfer (T^{unt})	household income	none (lump sum)	T^{unt}
Green subsidy (s_g)	switching cost ψ_g	number of switchers	$s_g \psi_g D^{\text{sw}}$
Carbon tax (τ_b)	brown price P_B^E , permanent	brown energy use	rebated (App. A)

Notes: The first four are crisis instruments, zero in the baseline (Section 6); the carbon tax is permanent and defines the prepared economy (Section 7). Its revenue is rebated lump sum in the baseline, with green-subsidy recycling in Appendix A.

The distinction that matters is between acting on the *price* and acting on *income*. The price cap holds P_B^E down through (18); because P_B^E is the price the adoption margin responds to (equation (11)), capping the price switches the margin off, and the cap's cost is a marginal subsidy: it scales with actual crisis energy use. The two transfers instead leave P_B^E free and support income, so the margin survives. They differ in their base: the *Slutsky* transfer (intensity ins_E) is indexed to each household's pre-crisis brown-energy use, so a switcher keeps the same cheque and the margin is undistorted, but, as Section 6 shows, indexing to energy use is regressive; the *flat* transfer T^{unt} is the equal-cost, unconditional alternative. The *green subsidy* pays a fraction s_g of the switching cost, forcing adoption directly rather than through the price. The *carbon tax* raises P_B^E permanently, greening the economy before any crisis; it is the ex-ante counterpart to the crisis instruments and the subject of Section 7.

All crisis outlays are deficit-financed. Collecting them term by term, with each the cost line of Table 2,

$$X_t = \underbrace{T_t^{\text{unt}}}_{\text{flat transfer}} + \underbrace{ins_E(p_t^E - \bar{p}^E) \bar{C}_E^B}_{\text{Slutsky transfer}} + \underbrace{\tau^E(p_t^E - \bar{p}^E) c_{E,t}}_{\text{price cap}} + \underbrace{s_g \psi_g D_t^{\text{sw}}}_{\text{green subsidy}}, \quad (28)$$

debt is stabilised by the distortionary labour tax,

$$B_t = (1 + r_{t-1}^{\text{ante}}) B_{t-1} + X_t - \tau_t^L w_t n_t, \quad \tau_t^L = \psi_B (B_{t-1} - \bar{B}), \quad (29)$$

with no debt in steady state ($\bar{B} = 0$): crisis spending is repaid by raising the labour tax in proportion to the debt it creates.

3.6 Equilibrium

Numeraire. The unit of account is the CPI basket, so $p^{\text{num}} = 1$ and the household budget is stated in real terms; P is also the index the monetary rule targets.

Market clearing. Goods, assets, and energy clear:

$$c_{H,t} + c_{H,t}^* - y_t = 0, \quad A_t - nfa_t - j_t - B_t = 0, \quad \begin{cases} P_t^{E*} \text{ exogenous} & (\text{elastic supply}), \\ c_{E,t} = E_t^s & (\text{fixed quantity}), \end{cases} \quad (30)$$

where the asset-market condition is redundant by Walras' law: it holds once goods, energy, and the government budget balance, so we do not impose it separately. The two energy closures correspond to the two experiments of Section 4.2: a price-taking economy facing an exogenous world-price path, and a fixed-quantity economy whose price clears an inelastic supply. Under the fixed-quantity closure the relevant market is the European one: availability is set region-wide and P^{E*} is the price that clears it against regional demand.

Let Ψ_t denote the joint distribution over the household state (d, e, a) , durable technology, idiosyncratic productivity, and assets, at date t . Taking as given the sequences of prices, taxes, and transfers, each household solves the recursive program of Section 3.1.3: it chooses nondurable consumption and savings by (7)–(8) and, at the durable stage, switches from brown to green with the logit probability (10). This program delivers value functions V_t and policy functions $\{c_t, a_t, P_t^{sw}\}(d, e, a)$. The distribution then evolves as $\Psi_{t+1} = \Lambda_t(\Psi_t)$, where the law of motion Λ_t is induced by these policies together with the exogenous productivity chain, and aggregates are the corresponding moments of Ψ_t (12).

Definition (competitive equilibrium). Given the world paths $\{P^{E*}, r^*, P^{F*}\}$, the initial distribution Ψ_0 and net foreign assets, and the monetary and fiscal rules, a competitive equilibrium is a sequence of aggregate prices $\{w, \pi, \pi^w, \pi^F, \pi^E, Q, p^E, r, r^{\text{ante}}, p_{H/P}\}_t$, value and policy functions $\{V_t, c_t, a_t, P_t^{sw}\}$, distributions $\{\Psi_t\}$, aggregate quantities $\{y, n, c_H, c_F, c_E, C, A\}_t$, technology shares $\{D^G, D^{\text{sw}}\}_t$, and fiscal variables $\{B, \tau^L\}_t$ such that:

- (i) (*Households*) given prices, taxes, and transfers, the value and policy functions solve the recursive problem (7)–(8), with brown-to-green switching governed by (10)–(11);
- (ii) (*Consistency*) Ψ_t evolves from Ψ_0 under the law of motion induced by the policy functions and the productivity process, and the aggregates in (12) are the moments of Ψ_t ;
- (iii) (*Firms and unions*) the price and wage Phillips curves (13)–(14) and the price system (15)–(19) hold;

- (iv) (*Government*) the monetary rule (27), the fiscal rule, and the government budget (29) hold;
- (v) (*Open economy*) uncovered interest parity (22), the balance-of-payments and net-foreign-asset accounting (25)–(26), and the energy-supply closure (23) hold;
- (vi) (*Clearing*) the goods and energy markets clear (30); the asset market clears by Walras’ law.

The object of interest is thus a sequence of prices and allocations, together with a distribution consistent with individual optimisation. The sequence-space Jacobian method of Auclert et al. (2021), applied in Section 3.7, computes the linearised transition of this equilibrium.

3.7 Steady state

Because the economy is a pure importer, the openness normalisations of Auclert et al. (2024) reduce to $\alpha_F = (\alpha - \alpha_E)/(1 - \alpha_E)$ and a trade elasticity $\gamma = \eta$ set to hit an import-demand target $\chi = 0.3$, with scale normalisations $y = 1$, $A^N = \mu$, $\alpha^* = \alpha$, and $\bar{E}^* = \alpha_E$. The steady state solves for the labour-disutility shifter ϕ , the top discount factor $\beta_{\max}$, output, and the switching cost ψ_g , against the wage Phillips curve, the net-foreign-asset condition, goods clearing, and the green-share target $\bar{D}^G = 5\%$. The switching cost is thus calibrated to the level of adoption, and Section 4 takes up the parameter the level cannot pin, σ_ε . The steady state is identical across all baseline experiments.

4 Calibration and experimental design

The aggregate block is inherited from Auclert et al. (2024); the durable/adoption block is this paper’s addition. We keep the aggregate calibration close to a generic energy importer: the AMRS parameters, an energy expenditure share of four percent, and a low intensive substitution elasticity. The durable stands for any household energy durable, a heat pump or an electric vehicle. The relevant economy is the European one of Sections 1 and 8; the aggregate calibration is an illustrative mapping onto it rather than a euro-area estimation. The steady state is identical across all baseline experiments. Table 3 collects the parameters.

Table 3: Calibration

Parameter	Value	Source / target
<i>Preferences and idiosyncratic risk</i>		
r real rate (quarterly)	0.01	Auclert et al. (2024)
σ inverse EIS	1 (log)	Auclert et al. (2024)
φ inverse Frisch	2	Frisch 0.5 (standard)
σ_e, ρ_e, n_e income process	0.57, 0.92, 7	Rouwenhorst; AMRS baseline
β types / spread / $\beta_{\max}$	3 / 0.06 / solved	mean MPC ≈ 0.30 ; $\beta_{\max}$ closes nfa
asset grid $[\underline{a}, \bar{a}]$, n_a	$[0, 400]$, 50	
<i>Consumption nest and openness</i>		
α_E energy expenditure share	0.04	Auclert et al. (2024)
η_E energy substitution	0.10	Auclert et al. (2024)
α total import share	0.30	Auclert et al. (2024)
α_F , $\gamma = \eta$	derived	import-demand target $\chi = 0.3$
<i>Firms, prices, wages</i>		
μ gross markup	1.03	Auclert et al. (2024)
θ_w wage Calvo	0.938	AMRS baseline
θ_E energy-price Calvo	0.65	AMRS baseline
θ_F import-price Calvo	0.90	AMRS baseline
ζ_{BG} real-wage stabilisation	5	AMRS baseline
<i>Energy supply and open economy</i>		
ε^E energy-supply closure	∞ / fixed $\bar{E}$	price-taking SOE / fixed-quantity (Bayer et al. (2026))
Γ stock arbitrage	100	AMRS baseline
$\bar{E}^{\text{shock}}$ world supply	α_E	endowment normalisation
<i>Monetary and fiscal</i>		
$(\phi_\pi, \phi_{\pi e})$ baseline / Taylor	(0, 1) / (1.5, 0)	constant real rate (AMRS) / Taylor
$\bar{B}$, ψ_B debt / feedback	0, 0.04	no SS debt; τ^Y solved
<i>Green adoption margin [this paper]</i>		
δ_g green breakdown rate	0.05	five-year durable life
$\varrho = \bar{P}_G^E / \bar{P}_B^E$	0.54	EV/ICE operating-cost ratio; App. E
ψ_g switching cost	0.770 (solved)	targets $\bar{D}^G = 0.05$
σ_ε logit taste scale	0.05	identified in Section 4.1
<i>Policy instruments (all 0 in the baseline)</i>		
τ^E price-cap intensity	$\{0, \dots, 1\}$	1 = full freeze
ins_E targeted transfer	$\{0, 1\}$	pre-crisis brown-energy indexed
T^{unt} untargeted transfer	$\{0, 1\}$	flat-transfer counterpart
s_g switching subsidy	$\{0, 1\}$	crisis / carbon-financed
<i>ETS (prepared economy only)</i>		
τ_b brown carbon tax	$\{0.10, 0.20, 0.30\}$	permanent, ψ_g fixed
τ_g green carbon tax	≈ 0	green near-clean
revenue recycling	rebate / green subsidy	Appendix A

Notes: The aggregate block is inherited from [Auclert et al. \(2024\)](#); the durable/adoption block is this paper's addition. ψ_g is solved to $\bar{D}^G = 5\%$; σ_ε is identified in Section 4.1.

The adoption block adds four parameters. The green durable depreciates at $\delta_g = 0.05$, a five-year life, which fixes the steady-state switching flow at $\bar{D}^{\text{sw}} = \delta_g \bar{D}^G$. We anchor the durable to the electric-vehicle margin, where the operating-cost gap is measured directly. The delivered green/brown ratio is the electricity cost of driving a battery-electric vehicle relative to the fuel cost of a comparable gasoline car, $\varrho = (P^G e_G)/(P^B e_B)$. At 2025 euro-area retail prices (electricity 0.29 EUR/kWh and gasoline 1.62 EUR/L, both inclusive of taxes) and real-world energy use of 0.21 kWh/km for the electric car and 7.0 L/100km for the gasoline car, this gives $\varrho = 0.54$. Tax-inclusive prices are the relevant ones here: the baseline carries no carbon tax, so the household compares retail prices, and the asymmetric taxation of the two fuels is part of the environment the crisis and the ETS act on. Appendix E lists every input and its source, and shows the policy ranking is invariant to ϱ across $[0.48, 0.80]$. The switching cost ψ_g is solved to hit a steady-state green share $\bar{D}^G = 5\%$. That leaves one free parameter, the logit taste scale σ_ε , which governs how responsive the margin is and which a level target cannot pin. Section 4.1 disciplines it.

4.1 Identifying the taste scale

The switching cost ψ_g and the taste scale σ_ε are not separately identified by the single steady-state target $\bar{D}^G = 5\%$. For any σ_ε , ψ_g re-solves to hit the same share, so the steady state is invariant along the $(\sigma_\varepsilon, \psi_g)$ locus (Table 13): the green share and the switcher mass do not move as σ_ε ranges over an order of magnitude and ψ_g with it, from 0.58 to 1.74.

The responsiveness of the margin, however, varies along the locus. The steady-state adoption elasticity, the response of the green share to a permanent improvement in the green/brown operating-cost ratio, at fixed ψ_g , falls from 4.4 at $\sigma_\varepsilon = 0.02$ to 0.8 at 0.20, and the crisis adoption wave falls with it, from a peak ΔD^G of 13.8 to 2.2 percentage points (Figure 27, in Appendix D). The adoption-channel magnitudes are therefore conditional on σ_ε : a level moment cannot discipline them, and a second moment, the adoption elasticity, is required.

The quasi-experimental vehicle literature provides that moment, on the upfront-cost margin. Gallagher and Muehlegger (2011) and Chandra et al. (2010) estimate that a \$1,000 tax incentive raises hybrid adoption by 31–38%, in a population of higher-income early adopters; Muehlegger and Rapson (2022) estimate a demand elasticity of -2.1 for electric vehicles among low- and middle-income households, roughly 8% per \$1,000 at their mean transaction price of \$26,000: the mass-market floor on responsiveness. In the model one unit is quarterly household consumption, taken as \$20,000, so the baseline $\psi_g = 0.770$ is a green premium of about \$15,400; a \$1,000 cut in ψ_g raises the switching flow by 39% at $\sigma_\varepsilon = 0.05$,

30% at 0.07, and 12% at 0.20 (Table 13).

Which end of the empirical range applies depends on who the marginal adopter is. At a 5% green share the model’s marginal adopters consume roughly twice the average brown household (Section 7): they are the higher-income early adopters of the hybrid-era estimates, not the mass-market population of Muehlegger and Rapson (2022), whose estimate therefore bounds responsiveness from below. Matching the early-adopter range pins $\sigma_\varepsilon \in [0.05, 0.07]$, with an implied green premium of \$15,000–\$18,000 within the heat-pump and EV range, whereas $\sigma_\varepsilon \geq 0.10$ would imply premia of \$22,000 and above. The baseline $\sigma_\varepsilon = 0.05$ is the elastic end of that interval, and the mass-market estimate is spanned by the sensitivity analysis. The operating-cost margin, the channel Beresteanu and Li (2011) document for gasoline prices and hybrid demand, then implies an elasticity of the green share to the energy-cost gap of 2.7 at the baseline. This responsiveness is a structural property of a rare margin rather than an artefact of the smoothing method; Appendix D develops the point.

The dollar mapping is the one assumption stated rather than estimated. A higher consumption unit lowers the model’s per-\$1,000 response and raises the implied premium, so the two criteria move together and the identified interval stays narrow. Halving the unit moves the interval toward higher σ_ε , doubling it toward lower; either way the interval lies inside the sensitivity grid of Table 13, over which the steady state and the policy ordering are invariant.

The aggregate results do not depend on any of this. The cumulative output loss $\sum y$ is flat across the locus (Table 13, last column), varying by under 4% while the adoption response varies tenfold, and the policy ordering follows from the relative price alone. The taste scale governs the size of the adoption channel; the cap-versus-transfer contrast does not depend on it.

4.2 The experiments

This subsection fixes what the rest of the paper holds constant: the two crisis experiments, the counterfactual that isolates the adoption channel, the policy instruments, and the welfare measure by which scenarios are ranked.

The two energy shocks. Two experiments drive the crisis, one per energy closure. The *price* shock doubles the world energy price on impact and lets it decay with a 16-quarter half-life under perfectly elastic supply,

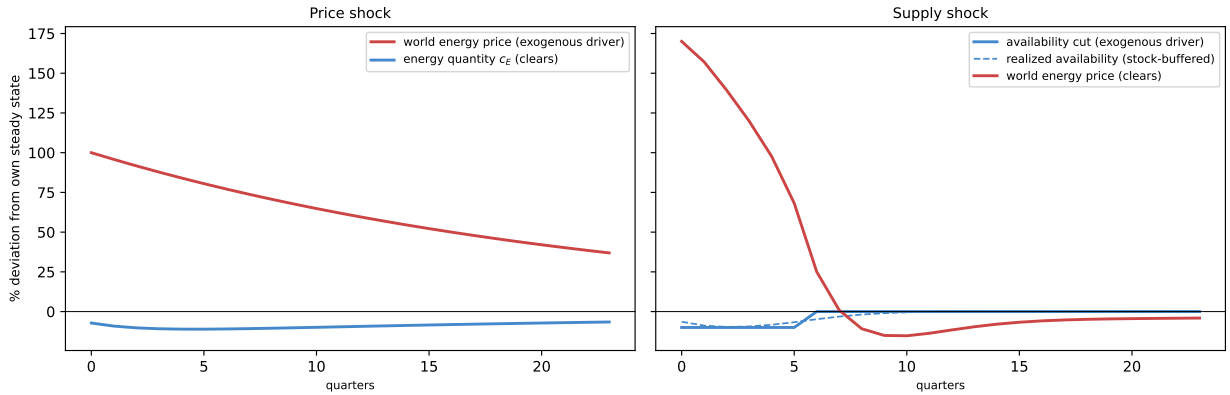
$$P_t^{E*} = \bar{P}^{E*} (1 + \nu \rho^t), \quad \rho = 2^{-1/h}, \quad (31)$$

with impact size $\nu = 1$, a doubling, and half-life $h = 16$ quarters, the experiment of [Auclert et al. \(2024\)](#). The *supply* shock removes 10% of the available quantity for six quarters and lets the price clear it,

$$E_t^* = \bar{E}^* (1 - 0.10 \cdot \mathbb{I}[t < 6]), \quad (32)$$

the fixed-quantity closure of [Bayer et al. \(2026\)](#). Figure 1 shows the two side by side. Under the price shock we move the world price directly and the quantity clears; under the supply shock we move the quantity, a 10% cut, and the clearing price adjusts. Because energy demand is highly inelastic ($\eta_E = 0.10$), a 10% quantity cut requires a price spike of about 170% to clear, so the supply shock is in fact the larger *price* disturbance of the two on impact. We report the price shock as the primary experiment. Under the fixed-quantity closure the clearing price must move to clear the market, and a policy that mutes the household demand response, such as the cap, forces it to move further still (Section 6); under the price shock the world price is exogenous, so the policy comparison is free of this price-clearing feedback. The supply shock is reported alongside it as the fixed-quantity counterpart. Section 5 puts the adoption margin to work under both.

Figure 1: The two energy shocks



Notes: Percent deviations from own steady state. *Left:* the price shock moves the world energy price (exogenous) and the quantity clears. *Right:* the supply shock moves availability (−10% for 6 quarters, exogenous) and the regional price clears the reduced quantity, spiking about 170% because demand is inelastic. In the right panel the bold blue line is announced availability and the dashed blue line is realised availability; they differ because world stock arbitrage (23) releases inventories against the cut and rebuilds them afterwards.

Isolating the adoption channel. Several results below isolate the adoption channel as the difference between the full model and a “no-adoption” counterfactual, $\Delta_H \equiv \sum_{t < H} (y_t^{\text{adopt}} - y_t^{\text{no-adopt}})$. The counterfactual is constructed to share the full model’s steady state *exactly*: households still hold the discrete durable-choice margin, and the steady-state green share, switching flow, and switching cost ($\bar{D}^G = 5\%$, D_{ss}^{sw} , ψ_g) are identical to the adoption econ-

omy. What differs is the *transition* only: the discrete-choice probabilities are held at their steady-state values throughout the impulse response, so the durable composition does not respond to the crisis while every other block responds normally. We call this the *common-steady-state*, or frozen-choice, counterfactual. The alternative, shutting the choice by making green durables infeasible, so the counterfactual’s own steady state has $D_{ss}^G = 0$, would compare two economies that differ both in the crisis response *and* in steady-state composition; the frozen construction isolates the crisis response alone.¹

The policy instruments. The instruments are those of Section 3.5 and Table 2. Four are crisis instruments, zero in the baseline and deployed once the shock has landed: the price cap τ^E acts on the brown retail price P_B^E ; the Slutsky transfer ins_E and the flat transfer T^{unt} act on income, the first indexed to pre-crisis brown-energy use and the second lump sum; the green subsidy s_g pays a fraction of the switching cost. All four are deficit-financed and repaid through the labour tax (29). The fifth, the permanent carbon tax τ_b , is the ex-ante counterpart: it raises P_B^E before any crisis and defines the “prepared” economy of Section 7, with its revenue rebated lump sum. Section 6 compares the crisis instruments and Section 7 the ex-ante one; the distinction that organises both is whether the instrument acts on the price or on income.

4.3 The welfare measure

We rank scenarios by the discounted utility of a household drawn at random before the shock. There are three permanent discount-factor types $i \in \{0, 1, 2\}$ of equal mass; type- i welfare in a scenario is

$$\mathcal{W}_i = \sum_{t \geq 0} \beta_i^t \left(\mathbb{E}_0[u(c_{it})] - v(n_t) \right), \quad (33)$$

the expectation over the date-0 (pre-shock) cross-section of type i ; the distribution then evolves endogenously along the transition, so (33) is the expected discounted utility of a household placed behind a date-0 veil of ignorance, the object a welfarist planner evaluates. Auclert et al. (2024), the model we extend, gauge the distributional effects of energy policy through consumption inequality; we complement that with a consumption-equivalent welfare ranking. We report $\mathcal{W}_i$ type by type and, as a population summary, the aggregate consumption-equivalent variation defined below. Felicity u is (2), evaluated on the bundle

¹Impulse responses are reported as 100× the level deviation from steady state. Variables whose steady state is one (prices, w , n , and the shares D^G , D^{sw}) read directly as percent or percentage-point deviations; C (steady state ≈ 0.999) is within rounding of a percent; y and gdp (steady state ≈ 0.998) are level deviations $\times 100$, within about 0.2% of the exact percentage; energy quantities are a share of the energy steady state (≈ 0.040).

the household consumes, so $\mathbb{E}_0[u(c_{it})]$ is the exact cross-sectional average, not felicity at the average bundle.

Labour is not a household choice. The union sets the wage (14) and hours n_t are common across households: a household of productivity e earns $e(1 - \tau^L)w n_t$ and works the same hours as everyone else. Preferences are separable and n_t is taken as given, so the labour-disutility term $v(n) = v_\varphi n^{1+\varphi}/(1 + \varphi)$ does not enter the household's consumption-saving problem (5). It is nonetheless a real utility cost of the hours worked, common to every type, so the welfare criterion must include it: we subtract $v(n_t)$ in (33). Omitting it would score a policy that raises employment as a free lunch: higher consumption with no offsetting disutility of hours.

From the transition to the CEV. Along a scenario's transition the felicity deviations are $\hat{u}_{it} = u_{it} - u_i^{\text{ss}}$, where $u_{it} = \mathbb{E}_t[u(c_{it})]$ is average type- i felicity at date t and u_i^{ss} its reference-steady-state level; the labour term deviates by $\Delta v_t = v'(n^{\text{ss}}) \hat{n}_t$, with $v'(n^{\text{ss}}) = v_\varphi (n^{\text{ss}})^\varphi$. The scenario's welfare relative to its own steady state is

$$\Delta \mathcal{W}_i = \sum_{t \geq 0} \beta_i^t (\hat{u}_{it} - \Delta v_t). \quad (34)$$

We express this in consumption units through the consumption-equivalent variation χ_i (Lucas, 1987): the permanent proportional consumption supplement that, granted to every type- i household in the reference steady state, reproduces $\Delta \mathcal{W}_i$. A permanent supplement scales felicity by $(1 + \chi_i)^{1-\sigma}$ and so raises welfare by $[(1 + \chi_i)^{1-\sigma} - 1] u_i^{\text{ss}}/(1 - \beta_i)$; equating to $\Delta \mathcal{W}_i$ and inverting,

$$\chi_i = \begin{cases} \exp[(1 - \beta_i) \Delta \mathcal{W}_i] - 1, & \sigma = 1, \\ \left[1 + (1 - \beta_i) \Delta \mathcal{W}_i / u_i^{\text{ss}} \right]^{1/(1-\sigma)} - 1, & \sigma \neq 1. \end{cases} \quad (35)$$

We report the per-type χ_i as the distributional object and, as the population figure, the aggregate consumption-equivalent variation χ^* : the common permanent supplement that, granted to every type in the reference steady state, reproduces the equal-weighted sum $\sum_i \Delta \mathcal{W}_i$,

$$\chi^* = \begin{cases} \exp[\sum_i \Delta \mathcal{W}_i / \sum_i (1 - \beta_i)^{-1}] - 1, & \sigma = 1, \\ \left[1 + \sum_i \Delta \mathcal{W}_i / \sum_i u_i^{\text{ss}} (1 - \beta_i)^{-1} \right]^{1/(1-\sigma)} - 1, & \sigma \neq 1. \end{cases} \quad (36)$$

The figure χ^* aggregates in common consumption units. A mean of the per-type χ_i would not: each χ_i is a share of a *different* type's consumption, and averaging such shares is not

a social-welfare aggregate (Dávila and Schaab, 2025). Both are in percent of permanent consumption; a negative value is a loss, the share of consumption a household would forgo in perpetuity to avoid the scenario.²

A common reference across steady states. Scenarios that begin from different steady states cannot be compared through (34) alone: the *ex-post* crisis instruments start at the pre-crisis baseline, whereas the *ex-ante* carbon tax has already moved the economy to a greener, permanently distorted steady state before the crisis. We fix the baseline steady state as the common reference and split each scenario’s welfare into a standing and a transition term,

$$\Delta\mathcal{W}_i = \underbrace{\frac{u_i^{\text{pre}} - u_i^{\text{base}} - \Delta v^{\text{lvl}}}{1 - \beta_i}}_{\text{standing gap vs. baseline}} + \underbrace{\sum_{t \geq 0} \beta_i^t (\hat{u}_{it} - \Delta v_t)}_{\text{crisis transition}}, \quad (37)$$

where u_i^{pre} is felicity at the scenario’s own pre-crisis steady state, u_i^{base} at the baseline, and $\Delta v^{\text{lvl}} = v'(n^{\text{base}})(n^{\text{pre}} - n^{\text{base}})$ the standing labour-disutility difference. The standing gap is zero for scenarios that start at the baseline and negative for the ex-ante tax, which gives up steady-state consumption to raise adoption; carrying it into $\Delta\mathcal{W}_i$ places ex-ante and ex-post policies on a single welfare axis. We feed (37) through (35) exactly as for any other scenario.

5 The crisis adoption wave

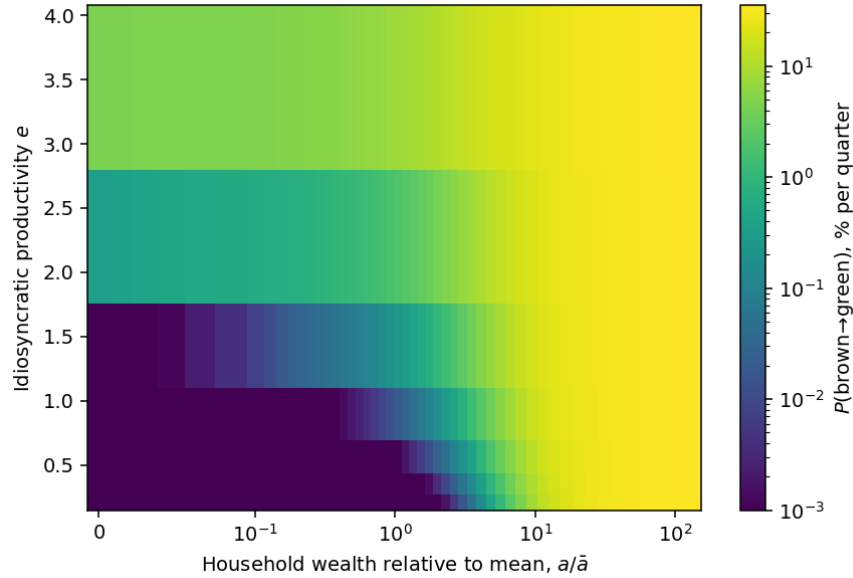
5.1 Who adopts at the steady state

At the calibrated steady state green durables are 5% of the stock and the quarterly switching flow is 0.26%, the flow that replaces green depreciation (Table 12, Appendix D). Adoption is far from uniform across households. Figure 2 plots the switching probability of the median discount-factor type over the wealth–productivity grid: the choice *policy function* rather than the mass-weighted pool, which mixes three types whose switching rates differ by orders of magnitude and is best read as the distribution in Table 12. The probability that a brown incumbent switches to green rises monotonically in both wealth and idiosyncratic productivity. Wealthier households can finance the switching cost ψ_g and more productive households expect the higher future income that makes the durable worth replacing, so

²The deviations in (34) are first order in the shock, so χ_i is the leading term of the welfare expansion around the (distorted) steady state along a deterministic transition. The setting is not the stochastic one in which a first-order evaluation can reverse welfare rankings (Kim and Kim, 2003; Benigno and Woodford, 2005): the transition is deterministic and the first-order term is nonzero and dominant. Because the shock is large and the adoption margin is convex, we bound the neglected second-order terms by recomputing every χ_i on the nonlinear transition (Appendix B.2); the levels move by at most about 5% and no ranking changes.

adoption concentrates among wealthy, high-productivity households. The surface also makes the rare-margin regime visible: away from that group the probability sits far below one. The brown mass occupies the exponential tail of the logit, where the semi-elasticity of switching is $1/\sigma_\epsilon$ household by household (Appendix D, Table 12).

Figure 2: Switching probability over the wealth–productivity grid

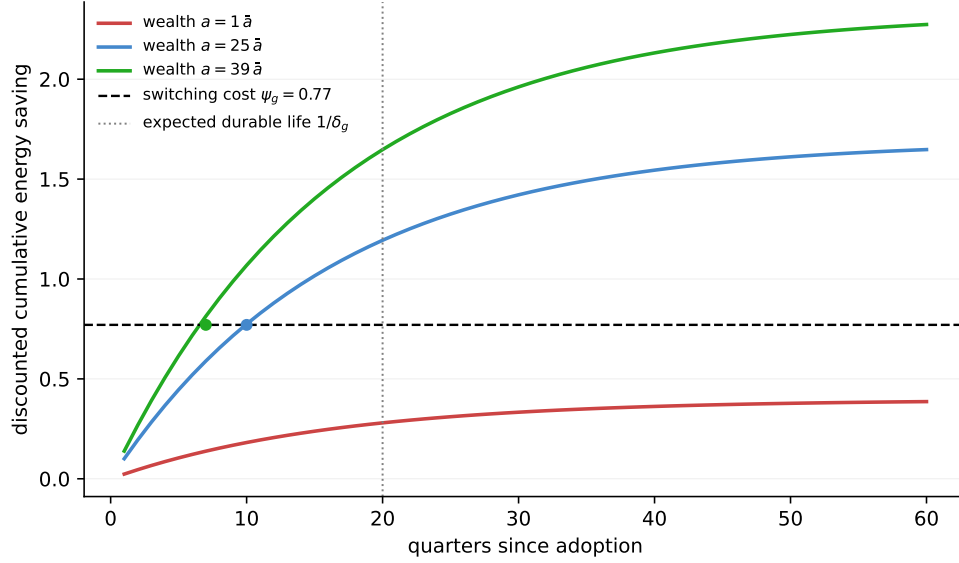


Notes: Probability of switching from a brown to a green durable, per quarter, for the median discount-factor type, as a function of household wealth relative to the mean $a/\bar{a}$ (horizontal, symmetric-log scale) and idiosyncratic productivity e (vertical), at the baseline steady state. Colour on a log scale. A single type's policy function is plotted rather than the mass-weighted pool: pooling the three discount-factor types, whose switching rates differ by orders of magnitude, makes the wealth gradient non-monotone even though each type's rule is monotone.

The wealth gradient is the marginal-switcher condition (11) seen household by household. Figure 3 evaluates one switch the way the condition does: the upfront cost ψ_g against the energy saving $(\bar{P}_B^E - \bar{P}_G^E) \bar{c}_E^G$ accumulated over the durable's life, discounted by interest and survival; the limit of the plotted sum is the discounted-saving side of (11). Because energy use rises with wealth in levels, so does the saving. At mean wealth the saving never reaches ψ_g : the typical household stays brown not because it discounts the future heavily but because its energy bill is too small for the operating saving ever to repay the cost. At about $25\bar{a}$ the breakeven arrives at ten quarters, half the durable's expected life; at $40\bar{a}$ it arrives within seven quarters, well inside the durable's life, and the switch clearly pays. The figure is an accounting cut: it holds the green incumbent's demand fixed and abstracts from consumption smoothing and the taste shocks that make the actual choice probabilistic. But it explains why the holding rate climbs from about 1.3% to 84% across the wealth distribution (Figure 4), and it is the object the crisis perturbs: a brown price shock scales

up the saving flow and moves each household's crossing left.

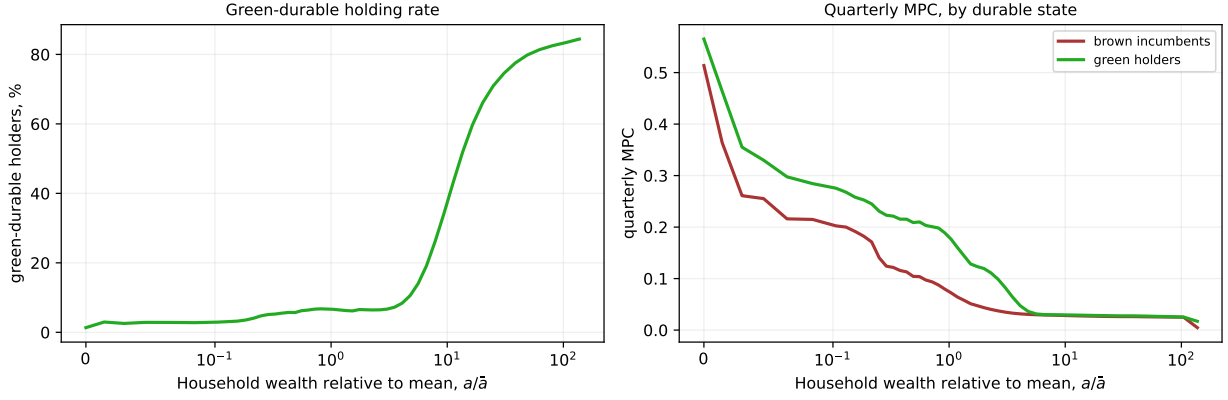
Figure 3: The private economics of one switch



Notes: Steady state, no shock. For the median discount-factor type at median productivity and three wealth levels: cumulative energy saving from operating green, discounted by interest and survival, $S(T) = \sum_{t \leq T} \left[(1 - \delta_g)/(1 + r) \right]^t (\bar{P}_B^E - \bar{P}_G^E) \bar{c}_E^G(e, a)$, against the switching cost ψ_g (dashed) and the expected durable life $1/\delta_g$ (dotted). Dots mark the breakeven quarter.

Two further steady-state objects anticipate the policy results (Figure 4). The switching *flow* has a stock counterpart: the green-holding rate rises from about 1.3% at the constraint to 84% at the top of the wealth grid, so the durable is held disproportionately by the wealthy. The marginal propensity to consume runs the other way, falling from about 0.5 at the constraint to near zero by $a \approx 3.5\bar{a}$, with brown incumbents and green holders close at each wealth level. These two gradients drive the incidence results of Section 6. A transfer to brown households reaches the high-MPC constrained tail and stabilises consumption effectively; an energy-indexed transfer does not, because the energy bill is flat as a share of spending (3.8% at the bottom of the wealth distribution against 3.5% at the top) and so rises with wealth in levels, sending the largest payments to the low-MPC wealthy. Homothetic energy demand makes energy-indexed targeting regressive; the flat share is the assumption behind that result, and the non-homothetic robustness check in Section 6 relaxes it.

Figure 4: Green-durable holding and MPC across wealth

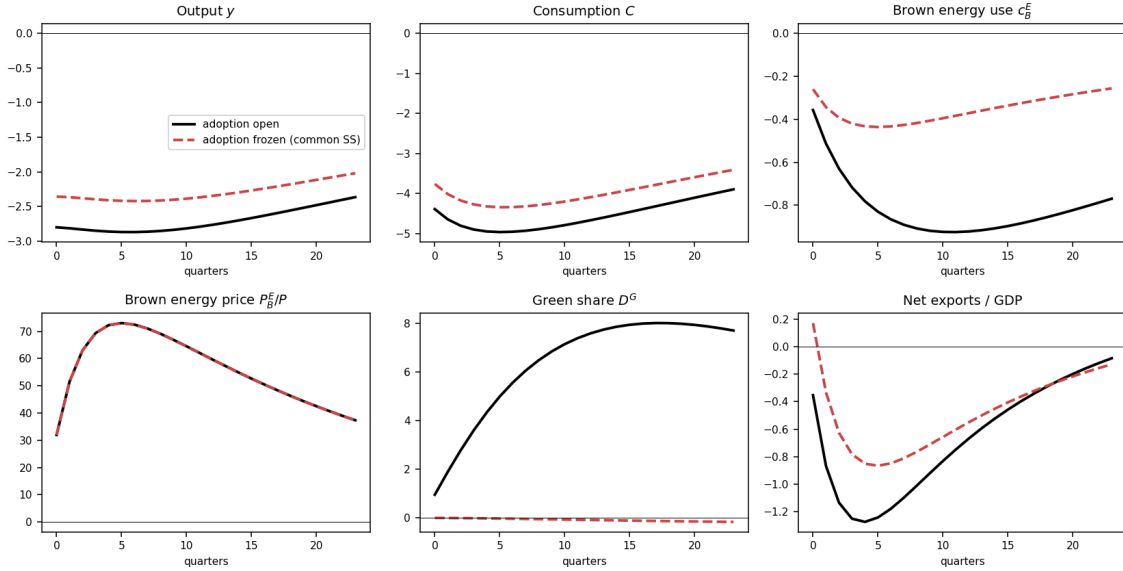


Notes: Aggregated over discount-factor types. Left: green-durable holding rate against wealth (the stock behind the switching flow). Right: quarterly MPC against wealth, by durable state. Wealth on a symmetric-log scale.

5.2 The baseline wave under the two shocks

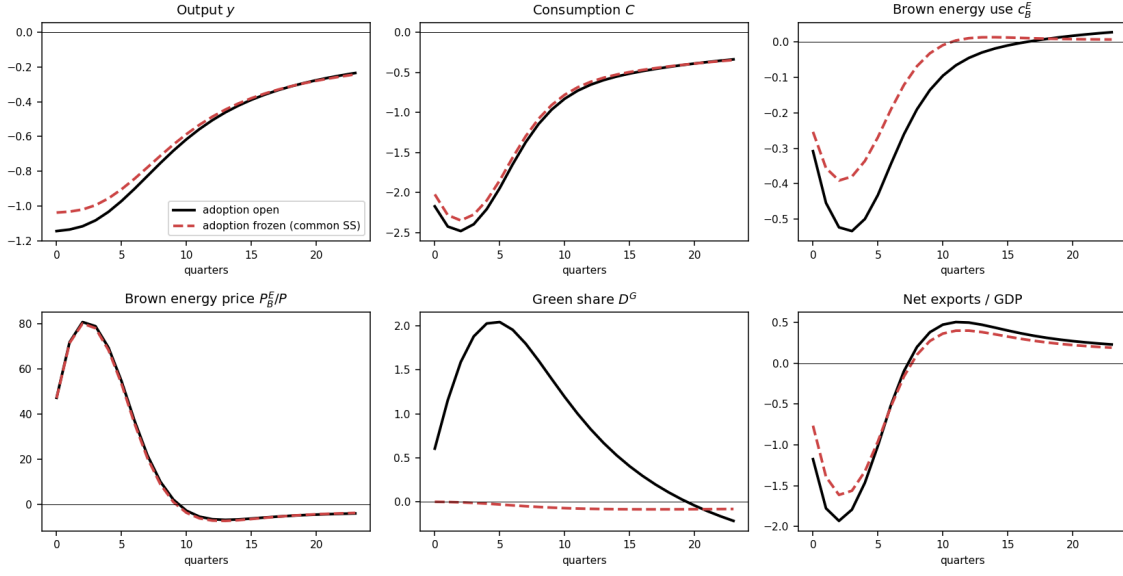
Figures 5 and 6 report the baseline responses to the two shocks, each with the adoption margin open (solid) against the frozen counterfactual (dashed).

Figure 5: Price shock: baseline response



Notes: Brown energy price shock, elastic supply, no policy. Adoption open (solid) against the frozen-choice counterfactual at the common steady state (dashed).

Figure 6: Supply shock: baseline response



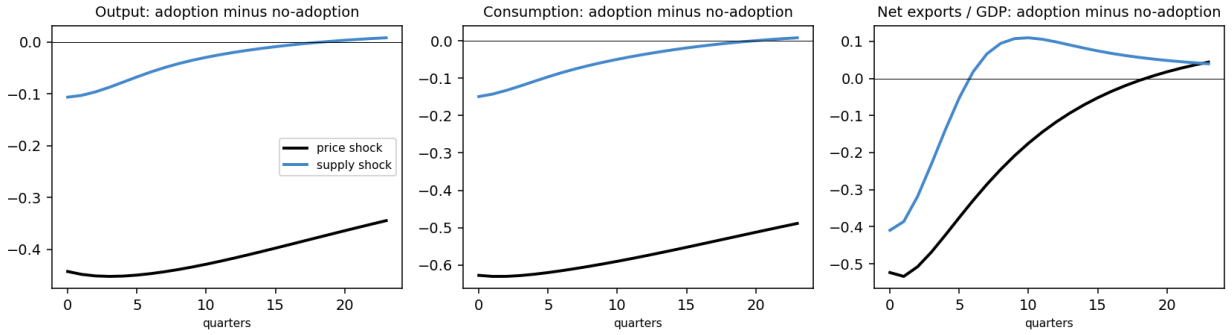
Notes: Brown energy supply shock (−10% for 6 quarters, fixed quantity), no policy. Adoption open (solid) against the frozen-choice counterfactual (dashed).

5.3 The adoption channel

Under the price shock, the adoption channel's contribution to cumulative output is $\Delta_{24} = -9.9$ (Table 4; Figure 7 plots the margin's contribution across variables): the margin is *contractionary*. It follows a race between two resource flows in the balance of payments (Section 3.4). Adoption imports a switching cost $\psi_g D_t^{sw}$, a resource outflow. It is paid once by each switcher, so it is front-loaded and scales with the number of adopters. The energy saving $(P_B^{E,pre} - P_G^{E,pre}) C_{E,t}^G$ works the other way: an inflow, booked as reduced imports, that accrues gradually over the life of the green durable. Under the frozen counterfactual the front-loaded cost dominates the windowed saving at every persistence. The force is the resource cost of the crisis-time switching wave at the calibrated $\psi_g = 0.77$ ($\approx 0.77 \times C_{ss}$) rather than a general-equilibrium artefact: the switching-cost outflow $\psi_g \times$ extra switchers cumulates to +11.74 on the import side, and net of the energy-mix inflow it leaves a net import draw of +4.98 (Figure 8) that pulls resources out of domestic absorption. Under the fixed-quantity (supply) closure a third, expansionary term appears: adoption lowers fossil demand against a fixed supply, so the clearing price falls and relaxes scarcity for *all* households, which shifts Δ_{24} up to −0.82. The marginal switcher's steady-state indifference, $\psi_g \approx (\bar{P}_B^E - \bar{P}_G^E) \bar{C}_E^G / (r + \delta_g)$, ties the threshold to the cost ratio ρ .

Booking caveat. What the booking convention moves is the sign of the adoption channels *contribution* to output, Δ_{24} , not the output response itself: aggregate output falls under both shocks and every booking. At the baseline $\alpha_F^{switch} = 1$ the switching cost and green bill are pure imports, so the contribution is negative and deepens the output fall; routing that spending onto the home good raises Δ_{24} , and under full domestic booking ($\alpha_F^{switch} = 0$) the contribution turns positive, +23.0 under the price shock and +0.58 under the supply shock. The policy ordering below is unaffected. We report $\alpha_F^{switch} = 1$ throughout and trace the full grid in Appendix A. This dependence is a booking convention rather than a result about the transition, and it is why we phrase the paper’s main result in terms of the *margin’s response* to policy rather than its output sign.

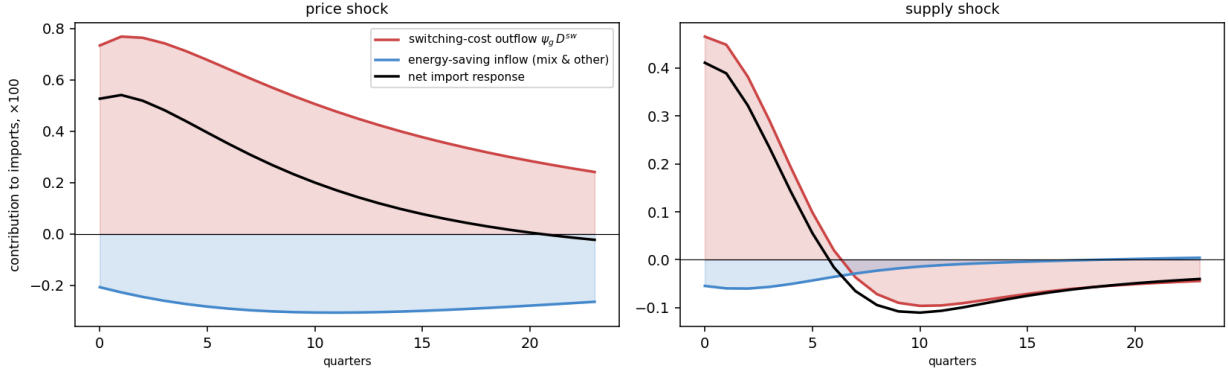
Figure 7: Contribution of the adoption margin, by shock



Notes: Full economy minus the frozen-choice counterfactual, by shock. Panels: output, consumption, net exports. Price shock in black, supply shock in blue.

The race between the two resource flows is exact on the import side. Figure 8 splits the adoption channel’s contribution to imports, full minus frozen, into the front-loaded switching-cost outflow $\psi_g D_t^{sw}$ and everything else, the latter dominated by the gradual energy saving from the brown-to-green mix shift. The split is exact by construction. The switching cost dominates the saving at every horizon under both shocks: under the price shock it cumulates to +11.74 against a -6.76 energy-mix inflow, a net import response of +4.98 that accounts for the channel’s contractionary output contribution ($\Delta_{24} = -9.9$). The outflow peaks on impact, when household income is falling hardest, and this timing is why booking it as an import makes the channel contractionary.

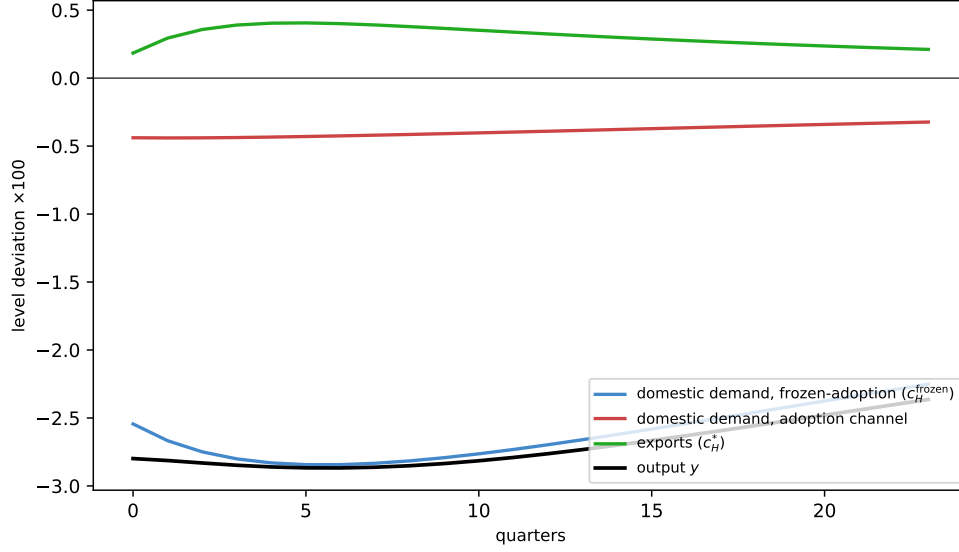
Figure 8: Adoption channel on the import side, by shock



Notes: Contribution of the adoption margin to imports, full economy minus the frozen-choice counterfactual. The figure plots the front-loaded switching-cost outflow $\psi_g D^{sw}$, the gradual energy-saving inflow, and their sum. Level deviations $\times 100$.

The adoption channel's output effect is a domestic-demand phenomenon. Decomposing output one level below the domestic/export split (Figure 9), the frozen-choice counterfactual separates output into three exact pieces: exports, domestic demand with the margin frozen (the AMRS real-income and substitution channels), and the adoption channel's increment to domestic demand. Under the price shock these cumulate to $y = -64.9 = (+7.4)_{\text{exports}} + (-62.9)_{\text{AMRS}} + (-9.4)_{\text{adoption}}$. The adoption drag (-9.4) falls entirely on domestic home-good demand: the switching-cost import outflow draws resources out of domestic absorption. It is close in magnitude and opposite in sign to the export gain $(+7.4)$, which is the AMRS competitiveness channel, so the two roughly cancel and aggregate output tracks the AMRS domestic channel (-62.9) .

Figure 9: Output response by channel



Notes: Price shock, no policy. Three exact channels: exports c_H^* , domestic demand with the adoption margin frozen (AMRS), and the adoption channel's increment to domestic demand. The three sum to y . Level deviations $\times 100$.

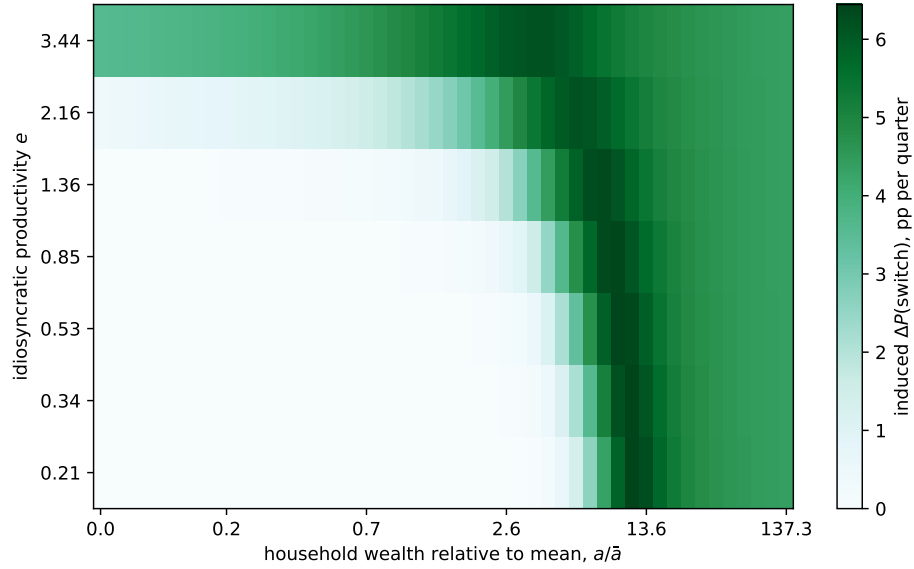
The same response, read from the household budget rather than from output, is almost entirely a general-equilibrium phenomenon. Decomposing consumption through the household consumption Jacobians, $dC = \sum_x J^{C,x} dx$ over the household's inputs and exact to first order, the direct energy-price effect is negligible (≈ 0 cumulative, energy being a small budget share) and the collapse is the indirect real-income channel (-97.9 of a -109.3 total, the real rate adding -3.5). The energy shock reaches consumption by lowering factor income in general equilibrium, not by raising the household's energy bill, the mechanism [Auclert et al. \(2024\)](#) and the broader HANK literature identify, here in an open economy with an adoption margin.

5.4 The induced adopters

The crisis draws in households who would not have switched at the pre-crisis steady state. Figure 10 characterises them with a comparative static: the switching probability of the median type at a steady state with a permanently 10% higher brown price, minus the baseline field. The induced switching is positive throughout and rises with productivity; at high productivity the induced margin reaches down to more moderate wealth, pulling in households who sat just below the switching threshold. Weighted by the steady-state distribution, the induced flow concentrates at moderate productivity and low-to-moderate wealth, where the brown mass sits. The construction is a comparative-static proxy: it uses

a permanent price change rather than the transitory crisis path, so it locates and signs the induced margin rather than timing it.

Figure 10: Induced adopters



Notes: Comparative static: the extra quarterly switching probability induced by a permanently 10% higher brown price, median discount-factor type, over the wealth–productivity grid. Contours mark the steady-state brown mass.

6 Ex-post shielding: prices versus incomes

The crisis instruments of the recent literature are a retail price cap and an income transfer. Both stabilise the crisis, and Table 4 shows they do so comparably. They differ on the margin this paper adds.

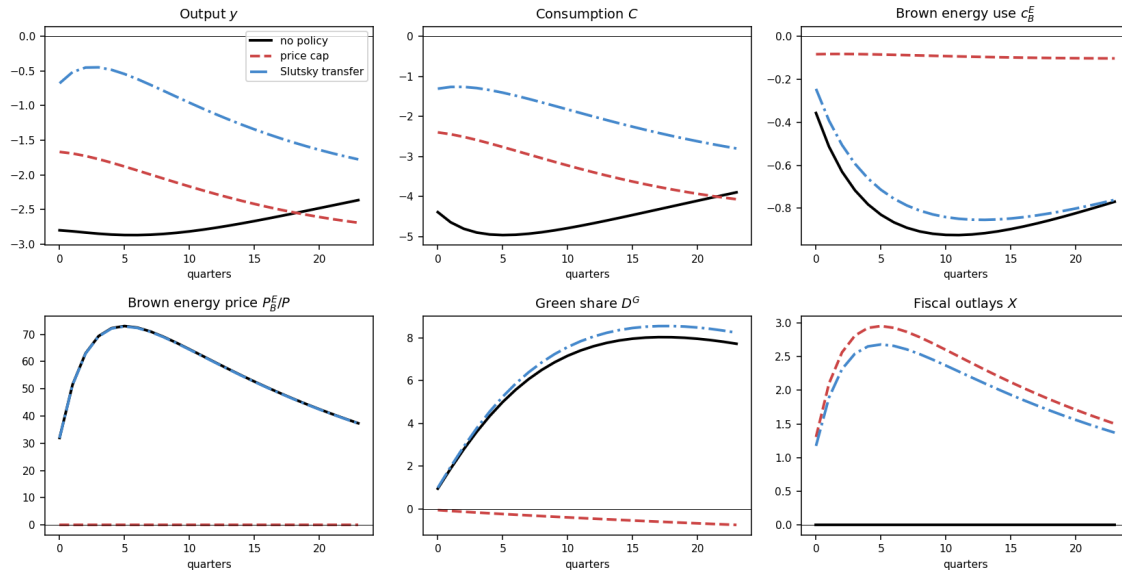
Table 4: Fiscal policy: summary

Shock	Policy	$y(0)\%$	$\sum y$	$\pi(0)$ ann.	peak D^G pp	fiscal cost	adoption $\rightarrow \sum y$
price	no policy	−2.80	−64.9	+5.56	8.03	0.00	−9.89
	price cap	−1.67	−53.1	−3.28	−0.05	54.11	+1.06
	Slutsky transfer	−0.68	−25.9	+7.28	8.54	49.19	−10.35
supply	no policy	−1.15	−14.9	+7.89	2.04	0.00	−0.82
	price cap	−1.71	−45.1	−1.36	−0.04	−19.43	+0.46
	Slutsky transfer	+0.43	−9.5	+10.84	2.83	18.03	−0.15

Notes: Cumulative sums over 24 quarters. The last column is the contribution of the adoption margin, the difference between the full model and a common-steady-state counterfactual in which the adoption choice does not respond to the shock (Section 3).

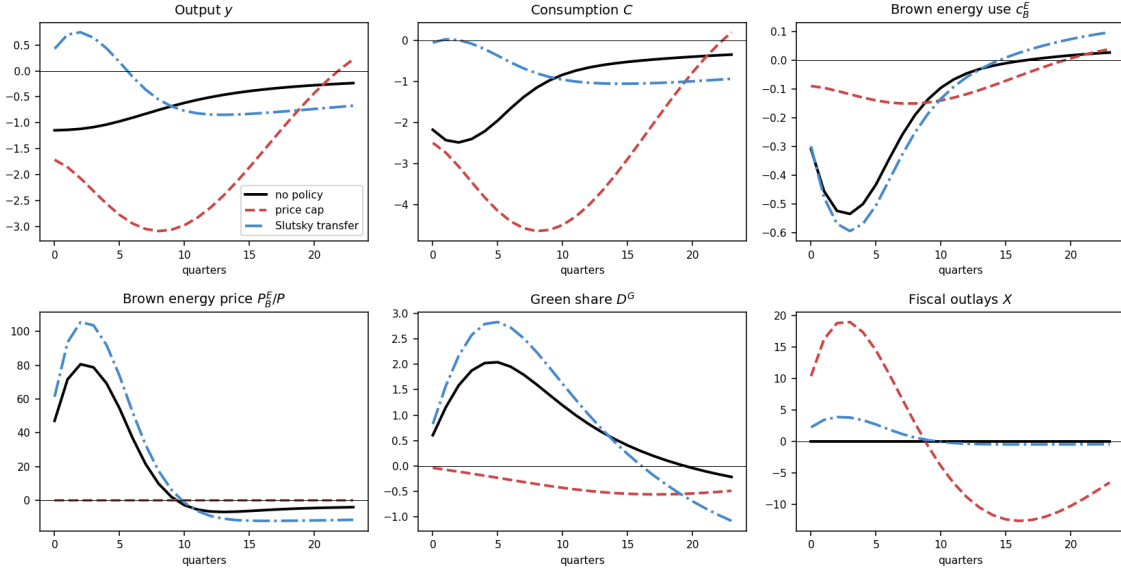
Cap versus transfer: the non-neutrality. Under the cap the peak green-share response falls from 8.0 pp to essentially zero (≈ -0.05 pp) under *both* shocks: the margin is not attenuated; it is switched off. The mechanism is direct: the cap holds P_B^E at its pre-crisis level, and P_B^E determines the return to adoption. The full fiscal responses under both shocks are in Figures 11 and 12 (no policy, price cap, Slutsky transfer).

Figure 11: Price shock: fiscal responses



Notes: Adoption open. No policy (solid), price cap (dashed), Slutsky transfer (dash-dotted).

Figure 12: Supply shock: fiscal responses



Notes: Adoption open. No policy (solid), price cap (dashed), Slutsky transfer (dash-dotted).

The Slutsky transfer leaves the green-share response essentially unchanged (8.5 pp against 8.0 with no policy) while delivering more output stabilisation than the cap at similar fiscal cost ($\sum y = -25.9$ vs. -53.1 ; gross fiscal 49.2 vs. 54.1). This preservation follows directly from the transfer's lump-sum indexing (Section 3.4): because it does not condition on the household's current durable, switching forfeits nothing and the relative-price incentive to adopt is left intact.

The sixth panel reports fiscal outlays X . Under the price shock the cap's outlay stays positive; under the supply shock it turns negative in the recovery (Figure 12). With the retail price frozen, household energy demand cannot adjust, so the fixed-quantity market clears through P^{E*} alone: the clearing price overshoots during the shock and falls below its pre-crisis level afterwards. The cap pays the gap between P^{E*} and the frozen retail price, so once P^{E*} drops below the freeze the scheme collects rather than pays. This reversal is larger than an integrated market's demand elasticity would produce, so the supply-shock cap magnitudes should be read with caution.

Two margins the frozen price governs. Adoption is not the only margin the frozen relative price governs. The open-economy identity $y = c_H + c_H^*$ splits home-good output into domestic demand for the home good and exports (Figure 13). Under no policy the energy shock lifts exports: the relative price of the home good falls and foreign demand rises, a partial offset to the domestic-demand collapse (cumulatively +7.4 against a domestic

−72.3). The cap switches that offset off almost entirely (+0.7): holding the retail energy price fixed suppresses the relative-price movement expenditure switching needs, exactly as it suppresses the one adoption needs. The transfer leaves the relative price free and preserves the export channel nearly intact (+6.1) while supporting domestic demand (−32.0). The cap and the transfer stabilise comparable amounts of output, but the cap does so by suppressing an adjustment margin and the transfer by supporting income: the same distinction the paper draws on the adoption share, read now through the trade balance.

Figure 13: Output response: domestic demand and exports



Notes: Brown-price shock, by policy. Output decomposed as $y = c_H + c_H^*$, domestic demand for the home good (blue) plus exports (green). Level deviations $\times 100$.

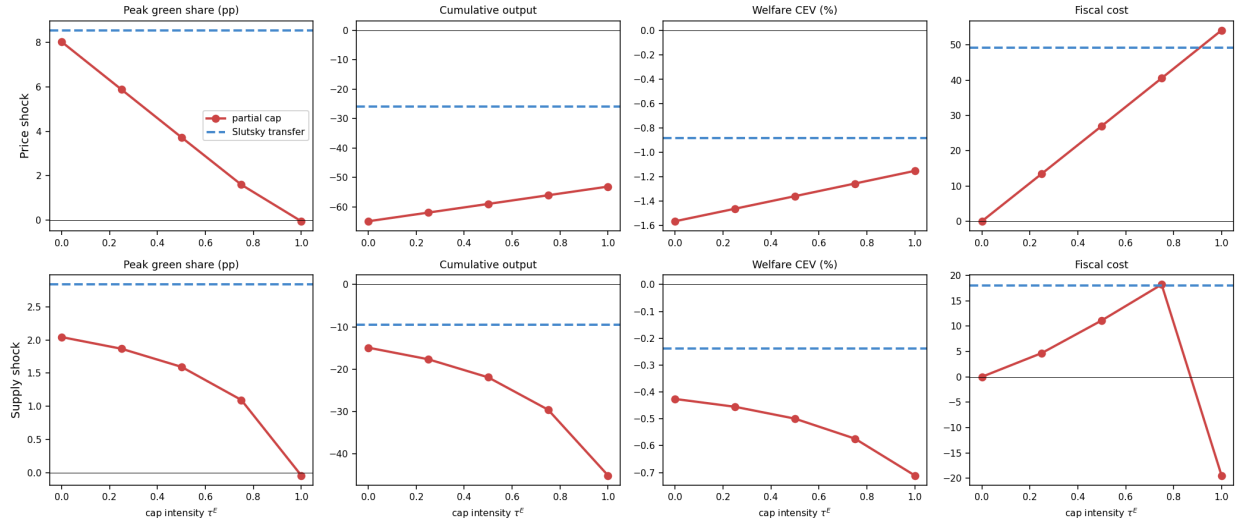
Partial caps. Table 4 uses $\tau^E = 1$, a complete price freeze; Table 5 sweeps $\tau^E \in [0, 1]$. Under the *price* shock the adoption loss is almost exactly linear in τ^E . Under the *supply* shock it is convex: small caps barely touch adoption and the collapse is concentrated near the full freeze. A partial shield is therefore much less damaging to the transition when supply is inelastic.

At the full freeze under the supply shock the fiscal cost is negative (Table 5), for the reason given in Section 6: in the recovery the clearing price falls below the frozen retail level, so the cap collects rather than pays.

Table 5: Price cap: response across intensity

Shock	Policy	peak D^G pp	$\sum y$	$\pi(0)$	$\sum c_E$	fiscal	CEV %
price	cap $\tau^E = 0.00$	8.025	-64.89	5.56	-8.65	0.00	-1.5662
	cap $\tau^E = 0.25$	5.877	-61.94	3.37	-7.30	13.48	-1.4636
	cap $\tau^E = 0.50$	3.728	-59.00	1.16	-5.94	26.99	-1.3605
	cap $\tau^E = 0.75$	1.594	-56.06	-1.05	-4.59	40.53	-1.2570
	cap $\tau^E = 1.00$	-0.050	-53.11	-3.28	-3.23	54.11	-1.1532
	transfer	8.543	-25.87	7.28	-6.11	49.19	-0.8828
supply	cap $\tau^E = 0.00$	2.043	-14.93	7.89	-2.41	0.00	-0.4260
	cap $\tau^E = 0.25$	1.867	-17.66	7.13	-2.41	4.69	-0.4552
	cap $\tau^E = 0.50$	1.595	-21.93	6.00	-2.42	11.11	-0.4993
	cap $\tau^E = 0.75$	1.097	-29.58	4.05	-2.44	18.19	-0.5736
	cap $\tau^E = 1.00$	-0.039	-45.11	-1.36	-2.80	-19.43	-0.7107
	transfer	2.834	-9.47	10.84	-2.41	18.03	-0.2373

Notes: Cap intensity τ^E swept from 0 to 1, against the Slutsky transfer benchmark (full compensation, $\iota = 1$). CEV is total consumption-equivalent variation.

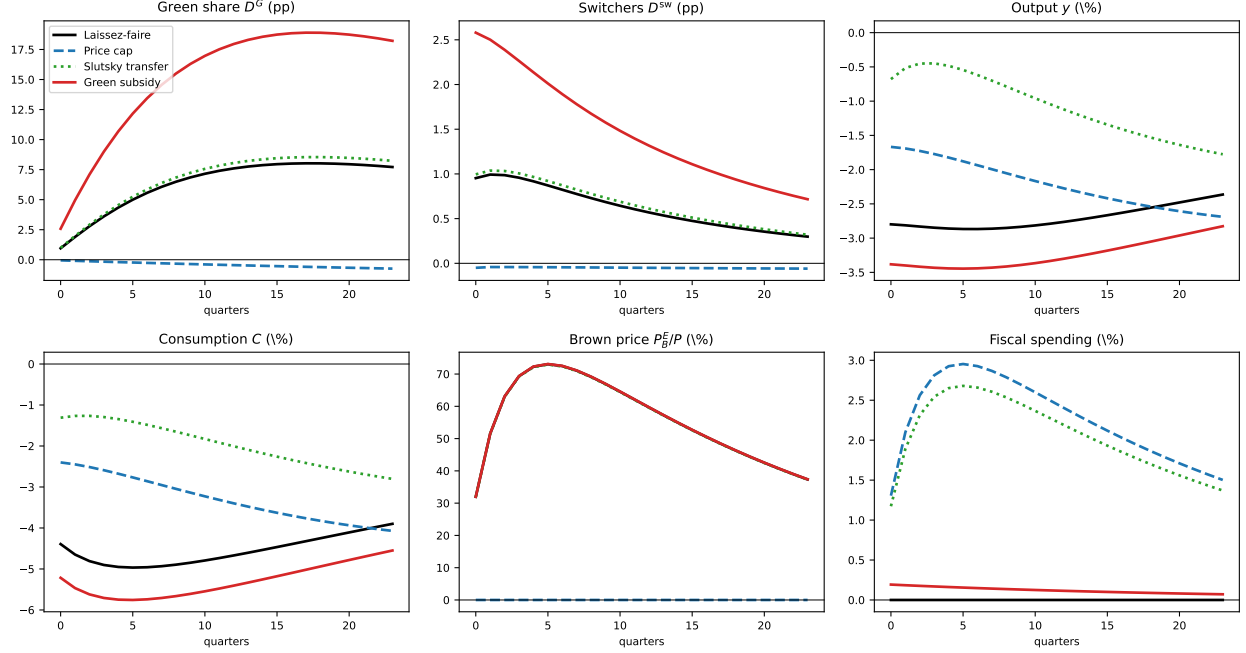
Figure 14: Price cap: response across intensity

Notes: Cap intensity τ^E (red) against the Slutsky transfer benchmark (blue dashed), for both shocks. Rows: price shock (top), supply shock (bottom).

The green subsidy. The green subsidy is a crisis instrument in its own right (Figure 15): the government pays a fraction of the switching cost during the acute crisis. It roughly

doubles the adoption response (peak D^G of 18.9 vs. 8.0 pp; switchers 2.6 vs. 1.0 pp) at trivial fiscal cost (2.9 against the cap's 54.1), yet its CEV is the worst in Table 9, -1.77% , below laissez-faire, and it deepens the output loss ($\sum y = -77.7$). Paying households to incur the real switching cost $\psi_g D^{sw}$, booked as an import (Section 3.4), in the middle of a consumption crisis destroys welfare even as it greens the economy.

Figure 15: The green subsidy



Notes: Brown-price shock, against laissez-faire, the price cap and the Slutsky transfer. The subsidy pays a fraction of the switching cost ψ_g during the acute crisis.

The subsidy here is a crisis instrument, paid from the deficit; its use as a recycling channel for permanent carbon revenue is a different exercise, taken up in Appendix A.2.

Who to shield: targeting and incidence. Without policy the impatient (high-MPC) households lose far more than the patient (Table 6). This gap makes the incidence of the transfer a first-order design question. The Slutsky transfer indexes the lump sum to each household's pre-crisis brown-energy use. Replacing it with an *untargeted* flat transfer of the same fiscal cost (49.2 in both, over $H = 24$) cuts the welfare loss by about a third, from a CEV of -0.88% to -0.58% (Table 9). It even improves output ($\sum y = -22.2$ vs. -25.9), while leaving adoption intact (8.5 pp). The gain is entirely distributional.

The mechanism is that energy-indexed targeting is regressive *in this model*. With homothetic CES energy demand the energy share is constant across the wealth distribution, so

brown-energy use rises with total consumption, hence with permanent income and patience. Indexing the transfer to pre-crisis energy use therefore hands the largest cheques to the low-MPC, wealthy households, where the insurance value per euro is lowest, while a flat transfer reaches the high-MPC households the crisis hits hardest. The result speaks to the 2022–23 European energy “brakes,” which were indexed to historical consumption: within the model the indexing is the source of the inefficiency rather than a refinement of it.

The result depends on the homothetic energy nest: if energy were a necessity with a declining Engel curve, the empirically standard case, low-income households would spend a *larger* share and the sign could reverse. The statement is conditional: *under constant energy shares, energy-indexed targeting is dominated by an equal-value flat transfer.*

Table 6: CEV by discount-factor type

Scenario (price shock)	CEV mean	impatient	middle	patient
no policy	−1.57	−2.41	−2.08	−1.21
cap $\tau^E = 0.5$	−1.36	−1.74	−1.67	−1.17
cap $\tau^E = 1$	−1.15	−1.05	−1.26	−1.14
transfer	−0.88	−0.70	−0.91	−0.91

Notes: Consumption-equivalent variation (%), private household welfare only. Types are the three permanent discount factors of Section 3, of equal mass.

7 Ex-ante carbon pricing

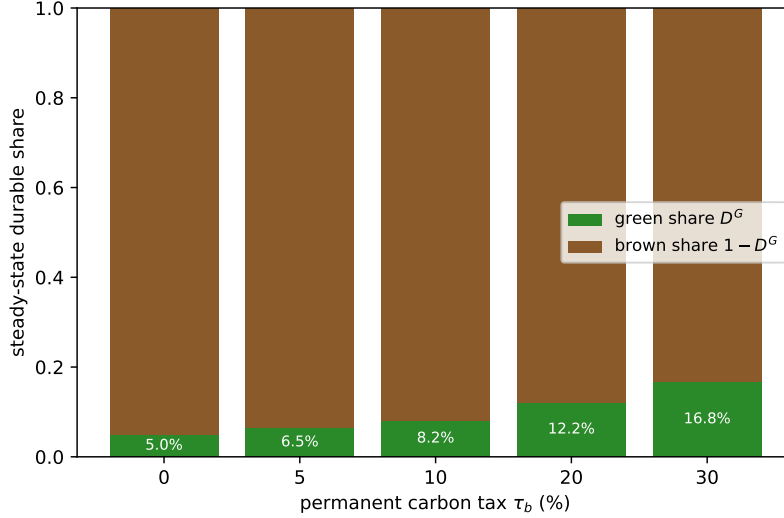
The instruments in Section 6 are all *ex post*: deployed once the shock has landed. Because the adoption margin responds to the same relative price the crisis moves, a natural alternative is to act *ex ante*, to lean on the margin in the steady state with a permanent carbon tax τ_b on brown energy, leaving the economy greener and possibly less exposed. This section evaluates that case. The answer is that ex-ante greening is a poor substitute for ex-post insurance, and the reason is intrinsic to the channel this paper adds: the adoption margin is an endogenous shock absorber, and pre-greening spends it.

7.1 A permanent carbon tax

A permanent carbon tax raises steady-state welfare in the model, through a second-best correction of the adoption friction rather than any externality. Holding ψ_g fixed at its no-tax value and letting D_{ss}^G float, the tax raises the effective brown/green price ratio and lifts steady-state adoption: D_{ss}^G moves from 5% at $\tau_b = 0$ to 8.2% at $\tau_b = 0.10$ and 16.8% at

$\tau_b = 0.30$ (Figure 16). Steady-state welfare rises monotonically in τ_b over the range the steady state solves: the standing CEV relative to the no-tax baseline is +0.28% at $\tau_b = 0.10$ and +0.56% at $\tau_b = 0.30$, with no interior optimum below $\tau_b = 0.30$, beyond which the steady state ceases to solve. The mechanism is second-best: green energy is structurally cheaper ($\bar{P}_G^E = \varrho < \bar{P}_B^E = 1$) but the logit taste scale σ_ε and the switching cost ψ_g leave green under-adopted, so taxing brown pushes the green share toward its efficient level. We treat this gain as a diagnostic of the channel rather than a policy recommendation: it would be dominated by any explicitly modelled externality or abatement cost. The carbon revenue is rebated lump-sum here; Appendix A.2 studies the budget-neutral alternative that spends it on a green switching subsidy, which greens the steady state *more* yet leaves a residual brown pool *less* responsive to the crisis: the same absorber logic as Section 7.3, across instruments.

Figure 16: Steady-state greening under a permanent carbon tax



Notes: ψ_g held fixed, carbon revenue rebated lump sum, no crisis. Steady-state durable composition: green share D_{ss}^G (green) and brown share $1 - D_{ss}^G$ (brown), across permanent carbon-tax levels τ_b .

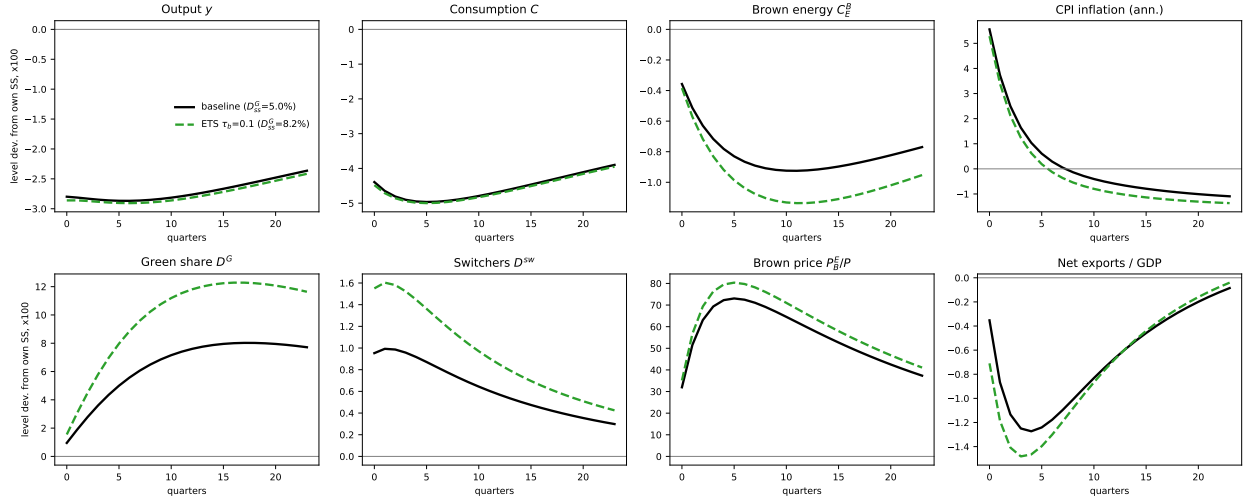
7.2 Ex-ante greening and crisis insurance

A steady state greened in advance by the carbon tax gives the economy almost no extra protection when the shock hits. We compare two economies under the *same* price shock: the baseline ($D_{ss}^G = 5\%$) and a “prepared” economy carrying a modest $\tau_b = 0.10$ ($D_{ss}^G = 8.2\%$). The prepared economy is essentially no better insured: its total CEV is -1.31% against -1.57% for laissez-faire, and that 0.25 gap is almost entirely the standing greening benefit (+0.28%) rather than crisis insurance; the crisis-only greening dividend is -0.03% , zero to first order (Section 7.3). Both are far worse than either ex-post instrument (cap -1.15% ,

transfer -0.88% ; Table 8). The greening the tax buys is real, D_{ss}^G is 64% higher, but too small a share of the economy to move aggregate exposure, and the tax's crisis-time value is negligible next to a transfer that reaches high-MPC households when the constraint binds. Consistent with Section 6, the combination “ETS + cap” tracks the cap alone: once the price is frozen the adoption margin is shut and the greener starting point is irrelevant.

Figure 17 overlays the two economies' responses to the same brown-price shock. The aggregates are nearly indistinguishable: output, consumption, inflation, net exports. The two differ visibly only in brown energy use, the green share D^G , the switching flow, and the brown price. The greener starting point changes the *composition* of the response rather than its aggregate size.

Figure 17: Baseline against prepared ETS under the same shock

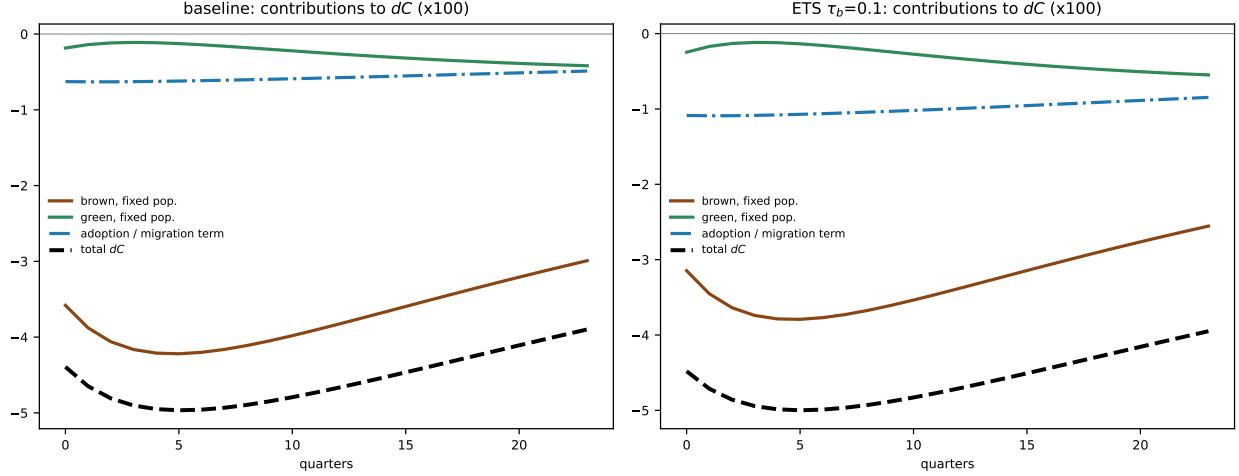


Notes: The same brown-price shock hitting the baseline ($D_{ss}^G = 5\%$) and the prepared ETS ($\tau_b = 0.10$, $D_{ss}^G = 8.2\%$) steady states, in level deviations from each economy's own steady state.

Figure 18 and Table 7 decompose the consumption response by the household's current technology. Switchers consume roughly twice the average brown household in the steady state, so a naive current-group split is contaminated by selection; we use the common-steady-state counterfactual to hold group populations fixed, measure the clean per-capita response of each group on the frozen path, and report the adoption/migration term (full minus frozen) separately. The aggregate loss is the same but the cross-section differs: under the ETS both brown and green incumbents lose *less* (per-capita -93.5% against -99.4% , and -27.3% against -37.3% of their own steady state over $H = 24$), and that relief is offset by a heavier adoption term (-23.8 against -13.7 , cumulative $\times 100$), more households paying ψ_g to switch during the crisis. Green incumbents, fully insulated from the brown price, still

lose about 1.5% per quarter, through the general-equilibrium income channel rather than energy.

Figure 18: Consumption by technology group, baseline against ETS



Notes: Brown-price shock, baseline (left) against prepared ETS ($\tau_b = 0.1$, right). Contributions to the aggregate consumption response dC : fixed-population brown, fixed-population green, and the adoption/migration term, with the total.

Table 7: Consumption response by technology group

Shock	Economy	$D_{ss}^G\%$	$\sum dC$	brown (fixed)	green (fixed)	adoption term	$\bar{c}_B\%$	$\bar{c}_G\%$	peak D^G
price	baseline	5.0	-109.3	-89.4	-6.1	-13.7	-99.4	-37.3	8.03
price	ETS $\tau_b = 0.1$	8.2	-110.3	-78.8	-7.7	-23.8	-93.5	-27.3	12.28
supply	baseline	5.0	-25.9	-21.6	-3.0	-1.2	-25.1	-8.6	2.04
supply	ETS $\tau_b = 0.1$	8.2	-26.5	-19.9	-4.1	-2.5	-25.2	-5.8	3.74

Notes: No policy, over $H = 24$. Cumulative level deviations $\times 100$: $dC = \text{brown (fixed population)} + \text{green (fixed population)} + \text{adoption term}$. The fixed-population terms come from the common-steady-state frozen-choice counterfactual and the adoption term is full minus frozen. $\bar{c}_j\%$ is the cumulative per-capita consumption response of brown and green incumbents, in % of their own steady state, at fixed populations.

Table 8: Ex-ante greening against ex-post capping

Scenario	CEV%	impat.	middle	patient	peak D_G	gross fisc.
Laissez-faire	-1.566	-2.407	-2.075	-1.210	8.03	0.00
Ex-post cap	-1.153	-1.055	-1.258	-1.137	-0.05	54.11
Ex-post transfer	-0.883	-0.702	-0.914	-0.910	8.54	49.19
Ex-ante ETS	-1.312	-2.093	-1.791	-0.978	12.28	8.28
Ex-ante ETS + cap	-0.863	-0.704	-0.936	-0.871	-0.08	62.04

Notes: Brown-price shock, $\tau_b = 0.1$. CEV is the total consumption-equivalent variation relative to the no-ETS steady state, by discount-factor type. Gross fiscal is the cap outlay (ex-post) or the standing carbon revenue over $H = 24$ (ex-ante).

7.3 Persistence

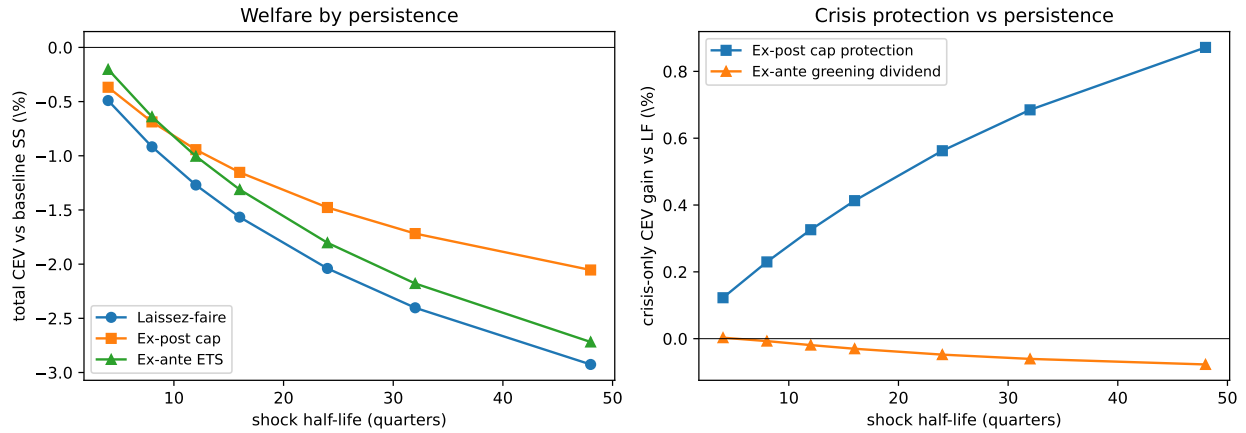
A natural defence of the ex-ante case is persistence: a longer-lived shock should let the greener economy's lower exposure compound over more periods. Figure 19 tests it by sweeping the shock half-life and re-solving only the impulse response (the steady state does not move with persistence), decomposing the ex-ante CEV into its constant standing part and a *crisis-only greening dividend*, defined as the prepared economy's crisis CEV minus the standing benefit minus laissez-faire.

The defence fails, and the failure is one of welfare rather than of switching volume. The crisis-only greening dividend does not grow with persistence; if anything it falls with the half-life of the shock. Its level is at most a few basis points at every horizon, inside the first-order approximation error of the CEV (Appendix B.2), so we do not sign it: it is zero to first order. The ex-post cap is different in kind: its crisis protection rises monotonically, from +0.26% to +1.33% as a longer freeze shields more of the transition, an order of magnitude larger than the approximation error and robust nonlinearly.

The reason is the adoption margin itself. A persistent brown-price shock makes switching attractive for many quarters, so the *browner* laissez-faire economy adopts heavily *during* the crisis and benefits from it over the whole episode; the margin is an endogenous shock absorber. The prepared economy has already spent much of that absorber in the steady state. It does not adopt less in the crisis; with a standing tax it in fact adopts *more* (peak D^G of 12.3 against 8.0 pp; Section 7.2). What it loses is the value of the marginal crisis switch: the households the tax has already moved to green would have switched anyway, so the crisis adds nothing for them, and the extra crisis adoption pays the real cost ψ_g at the trough (Section 6). The welfare-relevant absorber is the *option* to green cheaply exactly when the crisis makes it optimal, and the browner economy enters with that option more fully intact. Pre-greening raises the steady-state green share and even the crisis switching

volume, but not the crisis *welfare* value of the margin. The instruments that freeze the adoption technology, the cap here, and by construction the frozen-technology models of Bayer et al. (2026), Langot et al. (2026) and Auclert et al. (2024), miss an absorber that works best when the economy enters the crisis with room to green rather than already green.

Figure 19: Persistence and the two policies



Notes: Against the shock half-life. Left: total CEV of laissez-faire, the ex-post cap and the ex-ante ETS. Right: the crisis-only gain over laissez-faire, for the cap and for the ex-ante greening dividend.

8 The welfare ranking and European policy

The ex-post instruments of Section 6 and the ex-ante tax of Section 7 can now be placed on one welfare scale, and the ranking read against what conditions it.

The welfare ranking. Table 9 reports welfare, measured as the total consumption-equivalent variation (CEV) relative to the common no-tax steady state. On private welfare the Slutsky transfer dominates the cap under *both* shocks (price: -0.88% vs. -1.15% ; supply: -0.24% vs. -0.71%); the case for the transfer over the cap does not depend on the transition. But the *direction* of the cap's welfare effect depends on the supply elasticity: tightening the cap raises welfare under the price shock and lowers it under the supply shock (Figure 14).

Table 9: Price-shock policy responses

Scenario	$y(0)\%$	$\sum y\%$	peak D_G	gross fisc.	CEV%
Laissez-faire	-2.80	-64.9	8.03	0.0	-1.566
Ex-post cap	-1.67	-53.1	-0.05	54.1	-1.153
Slutsky transfer	-0.68	-25.9	8.54	49.2	-0.883
Flat transfer	-0.37	-22.2	8.45	49.2	-0.575
Green subsidy	-3.38	-77.7	18.90	2.9	-1.769
Ex-ante ETS ($\tau_b = 0.1$)	-2.86	-66.0	12.28	0.1	-1.312

Notes: $y(0)$ is impact output, $\sum y$ cumulative output over $H = 24$, peak D_G the peak green share, gross fiscal the total disbursement, and CEV the total consumption-equivalent variation relative to the no-ETS baseline steady state.

Reading the ranking. Three results point the same way. Energy-indexed targeting is dominated by a flat transfer of equal budget (Section 6); a permanent carbon tax is mildly welfare-improving in the steady state through a non-Pigouvian adoption correction, but essentially worthless as crisis insurance (Sections 7.1–7.2); and its insurance value is zero to first order and does not grow as shocks become persistent, because greening ex ante exhausts the adoption absorber (Section 7.3). The margin this paper adds is a stabiliser, not an instrument to activate before the crisis: it pays off when the economy is left free to adopt in the crisis.

Private welfare versus the transition. The CEV above is a *private* household-welfare measure: it values consumption and leisure along the transition and prices no climate or transition externality. Two results are therefore kept separate. On private welfare alone the transfer dominates the cap under both shocks; the case for the transfer does not depend on the transition. Separately, the transfer preserves the green adoption margin the cap eliminates (Section 6): a positive statement about the transition rather than a welfare ranking. We do not aggregate the two into one social-welfare number: that would require a social value of adoption (a carbon price times avoided brown-energy use) we do not model. A positive such value only reinforces the private ranking; it cannot reverse it, since the transfer already dominates.

The supply elasticity and the Langot–Bayer contrast. Under the supply shock the cap yields cumulative output of -45.1 against -14.9 with no policy: with inelastic supply, capping the price prevents substitution away from energy, so the economy cannot adjust its energy use at all. This result survives the addition of an adoption margin.

[Langot et al. \(2026\)](#), assuming elastic supply, find the French shield effective; [Bayer et al. \(2026\)](#), assuming fixed union-wide supply, find subsidies destructive. Our two closures reproduce the *sign* each obtains, which locates one source of the contrast in a single parameter. Once the supply elasticity is fixed, the direction of the cap’s welfare effect is determined, so part of the empirical disagreement reduces to measuring that elasticity at the relevant horizon.

The results map directly into the European policy architecture, on both the ex-post and the ex-ante side.

Shield incomes rather than prices. The €600-billion-plus shield of 2021–23 was dominated by price-suppressing measures, caps, regulated tariffs, energy tax cuts ([Bruegel, 2023](#)). In the model, price suppression delivers output stabilisation but sacrifices the entire crisis-time adoption wave: holding the brown retail price fixed removes the price signal that rewards switching, and the extensive margin shuts (Section 6). The data support the mechanism. European heat pump sales, having surged 38% to a record three million units in 2022 when gas was expensive, fell 5% in 2023 and 21% in 2024 as gas prices normalised, and the industry association’s own account names “the low price of subsidised gas” among the three main causes ([European Heat Pump Association, 2024, 2025](#)). A transfer of the same fiscal cost stabilises at least as well and leaves the wave intact. The 2022 episode left behind a fiscal bill and a forfeited adoption wave.

Pay the Social Climate Fund flat. The EU’s carbon price for buildings and road transport (ETS2), scheduled to start in 2028 after a one-year postponement, comes attached to a Social Climate Fund that will mobilise at least €86.7 billion over 2026–2032, of which up to 37.5% may finance temporary direct income support to vulnerable households ([European Commission, 2026](#)). Our distributional result speaks to the design of that component: because energy use tracks total consumption within technology groups, indexing support to energy expenditure targets wealth, not need. A flat payment of the same cost dominates the energy-indexed transfer in welfare, and the gain is concentrated on the constrained households the Fund is meant to protect (Section 6). Income support that preserves the price signal is the model’s best ex-post instrument; income support indexed to fossil-energy use is a weaker version of it.

Defend ETS2 as externality pricing rather than insurance. A recurring argument for pre-crisis carbon pricing is resilience: a greener economy, it is said, is better insured against the next fossil shock. The model disciplines that claim. The crisis dividend of the

prepared economy is zero to first order, shrinks with the size and persistence of the shock, and the mechanism is structural: pre-greening spends the adoption absorber, so the prepared economy enters the crisis with less welfare-relevant room to green, even though it switches more in volume (Section 7). The case for ETS2 is the externality and the standing welfare gain of the greener steady state (in the model a permanent brown tax raises standing welfare monotonically over the range we can solve, by about +0.28% at a 10% rate, through a non-Pigouvian adoption correction); it is not protection against the next crisis. The insurance rationale is the one the model supports least.

9 Conclusion

We added a single margin, an endogenous brown/green durable adoption choice, to an otherwise standard small open economy HANK model, and asked how crisis policy interacts with it. The central result is a non-neutrality: a retail price cap stabilises output but, by holding the brown price fixed, switches the green adoption margin off, whereas a transfer of the same fiscal cost delivers comparable stabilisation and leaves the margin intact. Shielding consumers from the relative price is not neutral for the energy transition, and Europe’s own heat-pump cycle (boom when gas was expensive, bust when subsidised gas made it cheap again) shows the margin operating in the data. The same logic, applied *ex ante*, yields the second result: a permanent carbon tax that greens the steady state provides no crisis insurance to first order, and the insurance value does not grow as the shock becomes persistent, because pre-greening spends an adoption absorber that works best when the economy enters the crisis with room to green. The case for carbon pricing is the externality; the case for crisis protection is income support, paid flat.

The model is built to isolate one margin, and its limitations indicate the next steps. The taste scale σ_ε is calibrated to the quasi-experimental vehicle-adoption literature rather than estimated on European durables; the identification in Section 4.1 makes the mapping explicit, and estimating the adoption elasticity directly is the next step; heat-pump sales against the electricity–gas price ratio across European countries would be the natural design. The switching cost is booked as an import by accounting necessity under the inherited markup structure; the ordering of the policy results does not depend on this convention, but the sign of the adoption channel’s output contribution does, and we have noted it as conditional throughout (Appendix A). The green side of the economy is an accounting construct rather than a technology: there is no green supply sector with capacity and investment, and no installer income distribution beyond a lump-sum rebate. The supply side more broadly has no capital, which likely makes the transfer somewhat more expansionary and the supply shock

more inflationary than they would be with an investment margin. Finally, energy demand is homothetic conditional on the durable type, so the model is silent on the energy-share-of-the-poor mechanism of [Bayer et al. \(2026\)](#); the two distributional channels complement each other.

None of these caveats changes what the paper establishes: the extensive margin of green adoption is a quantitatively important channel of energy-crisis policy, strong enough that the choice between shielding prices and shielding incomes is also a choice about the speed of the transition.

References

- G. Ascari, A. Colciago, T. Haber, and S. Wöhrmüller. Inequality along the european green transition. *The Economic Journal*, 2025. Article ueaf130.
- A. Auclert, B. Bardóczy, M. Rognlie, and L. Straub. Using the sequence-space jacobian to solve and estimate heterogeneous-agent models. *Econometrica*, 89(5):2375–2408, 2021.
- A. Auclert, H. Monnery, M. Rognlie, and L. Straub. Managing an energy shock: Fiscal and monetary policy. In *Heterogeneity in Macroeconomics: Implications for Monetary Policy*, volume 30 of *Series on Central Banking, Analysis, and Economic Policies*, pages 39–108. Central Bank of Chile, 2024.
- V. Audzei and I. Sutóris. A heterogeneous agent model of energy consumption and energy conservation. *Journal of Money, Credit and Banking*, 2026. doi: 10.1111/jmcb.70070.
- C. Bayer, A. Kriwoluzky, G. J. Müller, and F. Seyrich. Redistribution within and across borders: The fiscal response to an energy shock. *Journal of Political Economy Macroeconomics*, 2026. Forthcoming. Earlier version: “Hicks in HANK,” CEPR DP 18557.
- P. Benigno and M. Woodford. Inflation stabilization and welfare: The case of a distorted steady state. *Journal of the European Economic Association*, 3(6):1185–1236, 2005.
- A. Beresteanu and S. Li. Gasoline prices, government support, and the demand for hybrid vehicles in the united states. *International Economic Review*, 52(1):161–182, 2011.
- Bruegel. National policies to shield consumers from rising energy prices, 2023. Bruegel dataset, version of 26 June 2023.
- A. Chandra, S. Gulati, and M. Kandlikar. Green drivers or free riders? an analysis of tax rebates for hybrid vehicles. *Journal of Environmental Economics and Management*, 60(2): 78–93, 2010.
- F. Crucitti, M. Mersch, A. Michaelides, and A. Ostrovnaya. Green technology adoption over the life cycle, 2024. SSRN 5019631.
- E. Dávila and A. Schaab. Welfare assessments with heterogeneous individuals. *Journal of Political Economy*, 133(9), 2025. Earlier version: NBER Working Paper 30571.
- A. M. Dietrich, L. Leitenbacher, and G. J. Müller. Consumer durables, monetary policy, and the green transition. *European Economic Review*, page 105207, 2025.

- European Commission. Ets2: Buildings, road transport and additional sectors, 2026. Climate Action, climate.ec.europa.eu, accessed August 2026.
- European Heat Pump Association. Heat pumps in europe: Key facts and figures. Technical report, European Heat Pump Association, Brussels, May 2023.
- European Heat Pump Association. Pump it down: Why heat pump sales dropped in 2023. Technical report, European Heat Pump Association, Brussels, April 2024.
- European Heat Pump Association. Why did heat pump sales drop in 2024? Technical report, European Heat Pump Association, Brussels, March 2025.
- S. Fried, K. Novan, and W. B. Peterman. The distributional effects of a carbon tax on current and future generations. *Review of Economic Dynamics*, 30:30–46, 2018.
- S. Fried, K. Novan, and W. B. Peterman. Understanding the inequality and welfare impacts of carbon tax policies. *Journal of the Association of Environmental and Resource Economists*, 11(S1):S231–S260, 2024.
- S. Fried, D. Lagakos, and H. Rhodenhiser. Homework in climate economics: Household production, environmental preferences, and climate policy, 2025. Mimeo.
- K. S. Gallagher and E. Muehlegger. Giving green to get green? incentives and consumer adoption of hybrid vehicle technology. *Journal of Environmental Economics and Management*, 61(1):1–15, 2011.
- F. Iskhakov, T. H. Jørgensen, J. Rust, and B. Schjerning. The endogenous grid method for discrete-continuous dynamic choice models with (or without) taste shocks. *Quantitative Economics*, 8(2):317–365, 2017.
- J. Kim and S. H. Kim. Spurious welfare reversals in international business cycle models. *Journal of International Economics*, 60(2):471–500, 2003.
- D. Klenert, L. Mattauch, E. Combet, O. Edenhofer, C. Hepburn, R. Rafaty, and N. Stern. Making carbon pricing work for citizens. *Nature Climate Change*, 8(8):669–677, 2018.
- M. Kuhn and L. Schlattmann. Distributional consequences of climate policies. CEPR Discussion Paper 18893, CEPR, 2024.
- F. Langot, S. Malmberg, F. Tripier, and J.-O. Hairault. The macroeconomic and redistributive effects of the french tariff shield. *Journal of Political Economy Macroeconomics*, 2026. Forthcoming. Earlier version: CEPREMAP Working Paper 2305.

- A. Lanteri and A. A. Rampini. Financing the adoption of clean technology. Nber working paper, National Bureau of Economic Research, 2025.
- R. E. Lucas. *Models of Business Cycles*. Basil Blackwell, Oxford, 1987.
- E. Muehlegger and D. S. Rapson. Subsidizing low- and middle-income adoption of electric vehicles: Quasi-experimental evidence from california. *Journal of Public Economics*, 216, 2022.
- J. Rust. Optimal replacement of gmc bus engines: An empirical model of harold zurcher. *Econometrica*, 55(5):999–1033, 1987.

Appendix

The appendices collect supporting material. Appendix A derives the balance-of-payments booking of the adoption flows and shows how the booking convention signs the adoption channel. Appendix B reports the accuracy of the linearised solution and the nonlinear welfare check. Appendix C develops the monetary analysis: the adoption margin leaves monetary transmission unchanged, while the monetary stance moves the adoption wave only weakly relative to the fiscal instruments. Appendix D treats the logit taste scale as both a numerical smoothing device and a structural adoption elasticity, and reports its identification.

A Balance-of-payments booking of the adoption flows

The sign and magnitude of the adoption channel’s contribution to output depend on how the green-margin resource flows, the switching cost $\psi_g D^{sw}$ and the green energy bill, are booked in the balance of payments. The main text reports the pure-import baseline throughout. This appendix collects the two results that depend on that choice: the import-content sweep of the adoption spending, and the carbon-revenue recycling experiment, which is a booking-conditional statement about where the switching subsidy goes.

A.1 Import content of the adoption spending

The adoption cost enters the household budget as a CES bundle of home and foreign goods with import share α_F^{switch} . At $\alpha_F^{switch} = 1$ the switching cost is spent entirely on imports, the pure-import baseline of the main text; lowering it routes adoption spending onto the home good. The share is the continuous form of the import-versus-domestic booking of Section 3.4: the domestic booking of earlier drafts is the $\alpha_F^{switch} = 0$ endpoint, where neither the switching cost nor the green energy bill is a leakage.

The import content governs the sign of the adoption channel’s cumulative output contribution, Δ_{24} (full economy minus the frozen-choice counterfactual at the common steady state). Under the price shock it rises monotonically from -9.9 at $\alpha_F^{switch} = 1$, through $+6.3$ at $\alpha_F^{switch} = 0.5$, to $+23.0$ at $\alpha_F^{switch} = 0$: the leakage shrinks as spending shifts to the home good, and the channel turns expansionary once enough of it falls on the home good. Table 10 reports the grid for both shocks and Figure 20 the output paths.

Two invariances make clear that this is a booking statement rather than a statement about the crisis. Peak output stays negative across the whole grid, from -2.87% at $\alpha_F^{switch} = 1$ to -1.54% at $\alpha_F^{switch} = 0$: the import content dampens the recession; it never turns it into a boom. The peak green-share response is flat at roughly 8.0–8.2 pp throughout, so the

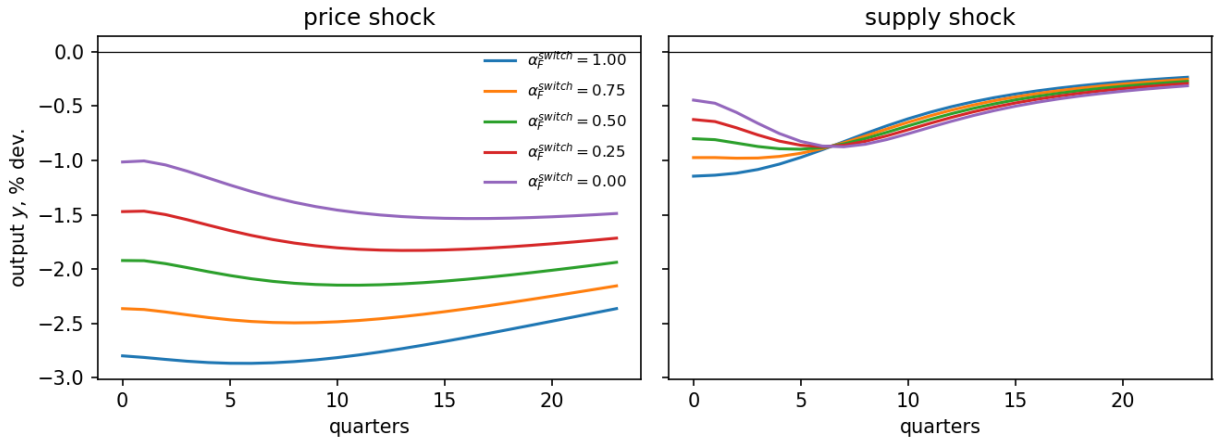
adoption margin itself is untouched by where its financing is booked. The policy ordering of Sections 6–7.1 is likewise unaffected. These invariances are why the paper states its main result in terms of the margin’s response to policy rather than the output sign of the channel.

Table 10: Adoption-spending import content

α_F^{switch}	shock	peak y (%)	peak ΔD^G (pp)	peak nx/gdp (%)	adoption $\rightarrow y$
1.00	price	-2.868	8.025	-1.273	-9.89
1.00	supply	-1.145	2.043	-1.931	-0.82
0.75	price	-2.495	8.096	-1.193	-1.91
0.75	supply	-0.980	2.052	-1.872	-0.45
0.50	price	-2.148	8.157	-1.112	6.26
0.50	supply	-0.897	2.060	-1.813	-0.09
0.25	price	-1.829	8.206	-1.031	14.57
0.25	supply	-0.874	2.066	-1.754	0.26
0.00	price	-1.535	8.243	-0.950	23.00
0.00	supply	-0.875	2.071	-1.693	0.58

Notes: Both shocks, across the import share α_F^{switch} of the switching-cost bundle. Columns: peak output, peak green-share response, peak net-exports-to-GDP, and the adoption channel’s cumulative output contribution Δ_{24} (adoption $\rightarrow y$).

Figure 20: Output response across the adoption-spending import share



Notes: Import share α_F^{switch} of the switching-cost bundle, price shock (left) and supply shock (right).

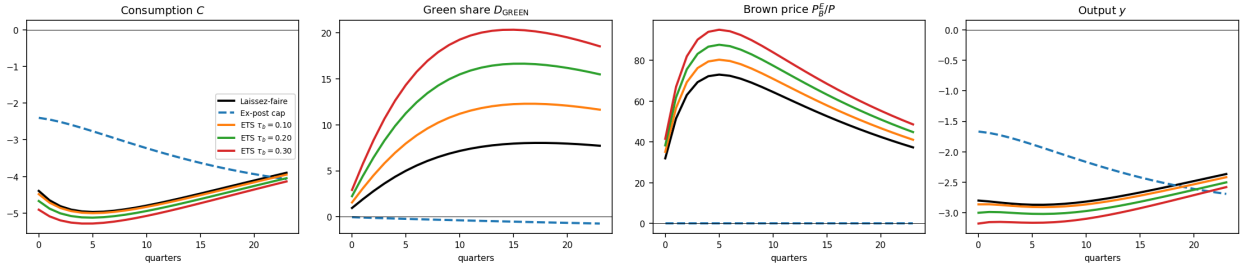
A.2 Recycling the carbon revenue: rebate versus green subsidy

The permanent carbon tax of Section 7.1 returns its revenue lump sum. An alternative spends it on the transition itself, paying a share s_g of the green switching cost. We compare

the two recyclings across carbon-tax rates $\tau_b \in \{0.10, 0.20, 0.30\}$, holding ψ_g at its no-tax value so the steady-state green share floats (Figures 21 and 22). Under the lump-sum rebate the steady-state green share rises from 8.2% to 16.8% across the grid, and the whole brown-price path in the crisis shifts up by the factor $(1 + \tau_b)$: the pre-tax price passes the world shock through identically, while households pay the tax-inclusive price. Under *budget-neutral* green-subsidy recycling, s_g solved so that the switching subsidy exactly exhausts the carbon revenue ($T^{\text{reb}} = 0$), the revenue funds a subsidy covering 11–18% of the switching cost and lifts the steady-state green share to 68–78%.

Two features stand out. First, the carbon base is self-eroding: at $\tau_b = 0.10$ the revenue is a third smaller under green-subsidy recycling than under the rebate, because the greener economy holds less brown-durable stock to tax. Second, the crisis-time greening response is *smaller* from the greener starting point under the subsidy, not larger: recycling that greens the economy through a lower switching *cost* preferentially converts the households nearest the logit margin, leaving a residual brown pool that is less responsive to the shock. The rebate does the reverse: it greens through a higher permanent brown *price* and leaves the residual pool closer to the margin. It is the adoption-absorber exhaustion of Section 7.3 seen across policy instruments rather than across shock persistence.

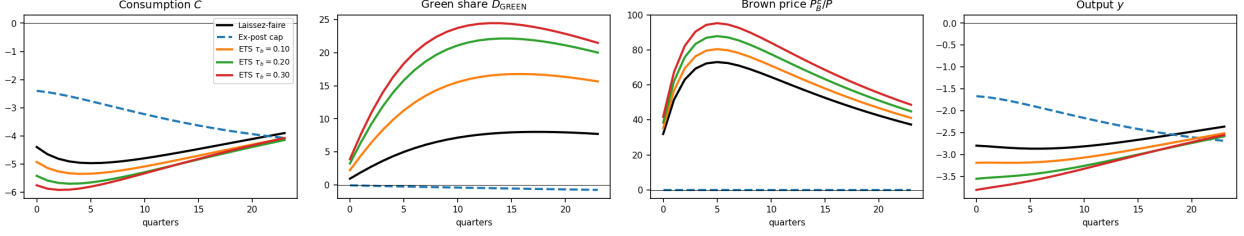
Figure 21: Carbon-tax sweep: lump-sum rebate



Notes: Laissez-faire and ex-post cap against the ex-ante ETS economy at $\tau_b \in \{0.10, 0.20, 0.30\}$, carbon revenue rebated lump sum.

Figure 22: Carbon-tax sweep: budget-neutral green-subsidy recycling

Version A: ex-ante ETS (τ_b sweep) vs ex-post cap under a brown-energy price shock -- budget-neutral green-subsidy recycling ($T^{\text{reb}} = 0$)



Notes: Budget-neutral recycling ($T^{\text{reb}} = 0$, s_g endogenous). Same axes as Figure 21.

B The linearised solution and the nonlinear welfare check

This appendix collects two robustness exhibits that support the main-text results without being part of the argument: the accuracy of the linearised (sequence-space) solution given the discrete adoption choice, and the first-order welfare check on the nonlinear transition.

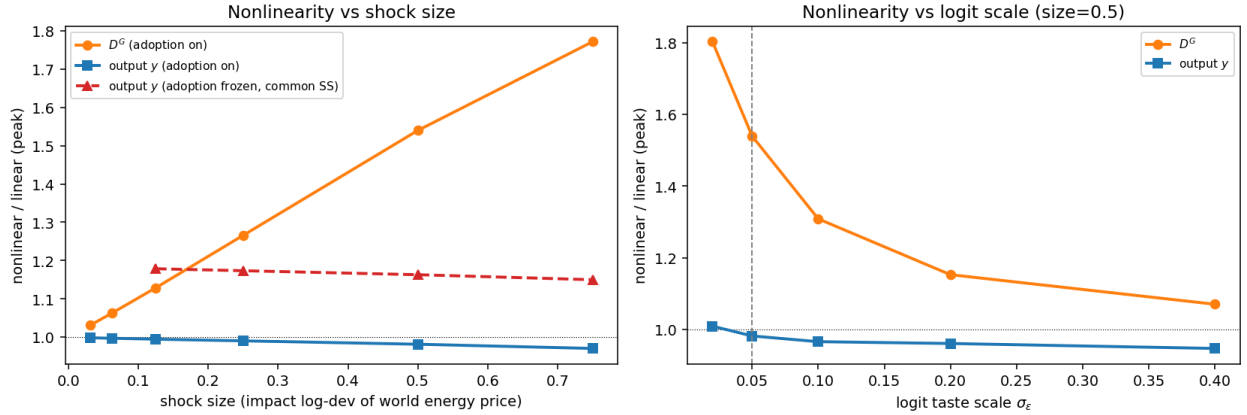
B.1 Accuracy of the linearised solution

A natural concern is that the discrete adoption choice makes the economy too nonlinear for the linearised (sequence-space) solution to be reliable, in contrast to [Auclert et al. \(2024\)](#), who find nonlinearity immaterial in the continuous economy. Comparing the linearised impulse responses with their nonlinear counterparts (a fixed-Jacobian Newton solve) confines the concern to one variable. Aggregate output is essentially linear: the ratio of the nonlinear to the linear peak response, $y^{\text{NL}}/y^{\text{L}}$, stays in $[0.97, 1.00]$ across shock sizes, a small nonlinearity comparable to what AMRS report. The nonlinearity is concentrated in the adoption share D^G , whose peak ratio rises monotonically from 1.13 at a one-eighth shock to 1.77 at a three-quarter shock (Figure 23, left). Freezing the margin does not reveal a more nonlinear economy underneath: under the common-steady-state counterfactual (Section 4.2), with D^G held fixed along the transition, output is itself only mildly nonlinear ($y^{\text{NL}}/y^{\text{L}} \approx 1.15$ at the largest shock), and freezing the margin widens the wedge between the nonlinear and linear paths rather than narrowing it. The discrete choice is where curvature concentrates, but leaving it free keeps aggregate output within three percent of its linear approximation.³

³At the baseline shock size ($\varepsilon = 1.0$) the nonlinear solution does not converge, so the sweep is reported to $\varepsilon = 0.75$. The D^G ratio is monotone and convex over $[0, 0.75]$, so nothing depends on the largest point.

The convexity is the intrinsic curvature of a logit share evaluated near a low level. For a binary choice $P = [1 + \exp(-\Delta/\sigma_\varepsilon)]^{-1}$ in the value gap Δ , the relative curvature is $(1 - 2P)/\sigma_\varepsilon$, so $D^{G,NL}/D^{G,L} - 1 \propto 1/\sigma_\varepsilon$. Re-solving on a grid of σ_ε (holding $\bar{D}^G = 0.05$ by recalibrating ψ_g) confirms this: the ratio falls monotonically from 1.80 at $\sigma_\varepsilon = 0.02$ to 1.07 at $\sigma_\varepsilon = 0.40$ (Figure 23, right), and it tends to one as the shock vanishes, so the linearisation is exact to first order. Two implications follow. Aggregate and consumption-based welfare statistics are reliable under linearisation. The *level* of induced greening at the largest shocks is understated by the linear solution, and does not converge under a full-size shock, so adoption-channel magnitudes are read at a reduced shock size. The taste scale σ_ε that governs this curvature is the same parameter Section 4.1 and Appendix D identify: an adoption-elasticity moment pins both at once.

Figure 23: Nonlinear against linear responses



Notes: Left: peak nonlinear/linear ratio against shock size, for output (adoption on and off) and D^G . Right: the same ratio against the logit taste scale σ_ε at a half-size shock, ψ_g recalibrated to $\bar{D}^G = 5\%$.

B.2 First-order welfare and the nonlinear check

The consumption-equivalent variations reported throughout are first-order in the shock: flow utility along the linear impulse response is a first-order deviation of aggregate felicity, so the CEV is the leading term of the welfare expansion around a distorted steady state along a deterministic path. The setting is not the stochastic one in which first-order welfare evaluation is known to give spurious rankings (Kim and Kim, 2003; Benigno and Woodford, 2005): the first-order term is nonzero and dominant here. But the shock is large and the adoption margin is convex, so the size of the neglected second-order terms is an empirical question, which we settle by recomputing each scenario's welfare on the nonlinear perfect-foresight transition (Table 11).

Table 11: Linear against nonlinear welfare

size	Scenario	CEV% (linear)	CEV% (nonlinear)	NL/L	impat.	middle	patient
0.25	Laissez-faire	-0.3939	-0.3911	0.993	0.987	0.992	0.996
0.25	Ex-post cap	-0.2895	-0.2871	0.992	0.986	0.989	0.994
0.25	Slutsky transfer	-0.2214	-0.2200	0.994	0.980	0.988	0.998
0.25	Flat transfer	-0.1439	-0.1444	1.003	0.887	1.006	1.002
0.25	Ex-ante ETS	-0.1169	-0.1175	1.005	1.006	1.008	1.001
0.5	Laissez-faire	-0.7862	-0.7736	0.984	0.977	0.983	0.988
0.5	Ex-post cap	-0.5783	-0.5697	0.985	0.974	0.981	0.989
0.5	Slutsky transfer	-0.4424	-0.4334	0.980	0.962	0.970	0.986
0.5	Flat transfer	-0.2877	-0.2862	0.995	0.799	0.990	0.993
0.5	Ex-ante ETS	-0.5168	-0.5136	0.994	1.007	1.002	0.981
0.75	Laissez-faire	-1.1770	-1.1420	0.970	0.967	0.971	0.971
0.75	Ex-post cap	-0.8661	n.c.	—	—	—	—
0.75	Slutsky transfer	-0.6628	-0.6323	0.954	0.935	0.940	0.962
0.75	Flat transfer	-0.4312	-0.4172	0.967	0.874	0.941	0.970
0.75	Ex-ante ETS	-0.9151	-0.8925	0.975	1.001	0.991	0.952

Notes: Brown-price shock at reduced sizes. CEV is the total consumption-equivalent variation relative to the no-tax baseline steady state; NL/L is the ratio of nonlinear to linear CEV, overall and by discount-factor type. The full shock (size 1.0) does not converge nonlinearly.

Three things follow. First, the linear CEV *overstates* the welfare loss, by 1–4% at a quarter-size shock and up to 7–26% at three-quarters, in the direction the convexity of the adoption margin predicts; the paper’s levels are upper bounds on the losses, and would be overstated by an extrapolated 10–15% at the full shock size, which does not converge nonlinearly. Second, the policy ordering, flat > Slutsky > cap > ETS > laissez-faire, holds at every converged size, linear and nonlinear alike, with the first-order gaps essentially unchanged; the welfare rankings stand. Third, the one difference in the other direction is the impatient type under the flat transfer, whose nonlinear/linear ratio rises to 1.89: the linear solution understates the gain of high-MPC households when the transfer relaxes a binding liquidity constraint, which reinforces rather than weakens the distributional result of Section 6. The single claim the first-order calculation cannot support is the sign of the crisis-only greening dividend of Section 7.3, which is why we report it there as zero to first order.

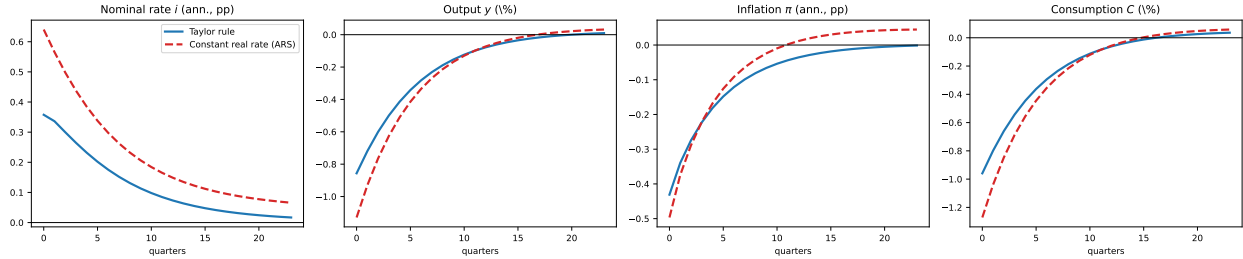
C Monetary policy and the adoption margin

Two questions connect monetary policy to the adoption margin, and they point in opposite directions. Does the margin reshape how a monetary shock transmits? And does the monetary stance, during an energy shock, reshape the adoption wave? The first is a robustness check; the second is where the interest-rate sensitivity of green durables that [Dietrich et al. \(2025\)](#) identify in a representative agent shows up in our setting.

C.1 Monetary transmission

The energy-shock experiments hold monetary policy at the AMRS constant-real-rate baseline. A 100-basis-point (annualised) tightening isolates the role of the rule itself. Figure 24 contrasts the constant-real-rate baseline with a standard Taylor rule $(\phi_\pi, \phi_{\pi e}) = (1.5, 0)$. The response is standard: output and consumption fall, inflation falls. It is *not* materially reshaped by the adoption margin. At a 5% green steady state the durable-choice channel is too small to move aggregate monetary transmission (on impact, output -0.86 under the Taylor rule against -1.13 under the constant-real-rate rule; inflation -0.43 against -0.50 annualised). The adoption margin matters for how the economy absorbs an *energy* shock, not for how it transmits a *monetary* one.

Figure 24: Monetary-policy shock



Notes: 100bp annualised tightening: Taylor rule (solid) against the constant-real-rate baseline (dashed). Panels: nominal rate, output, inflation, consumption.

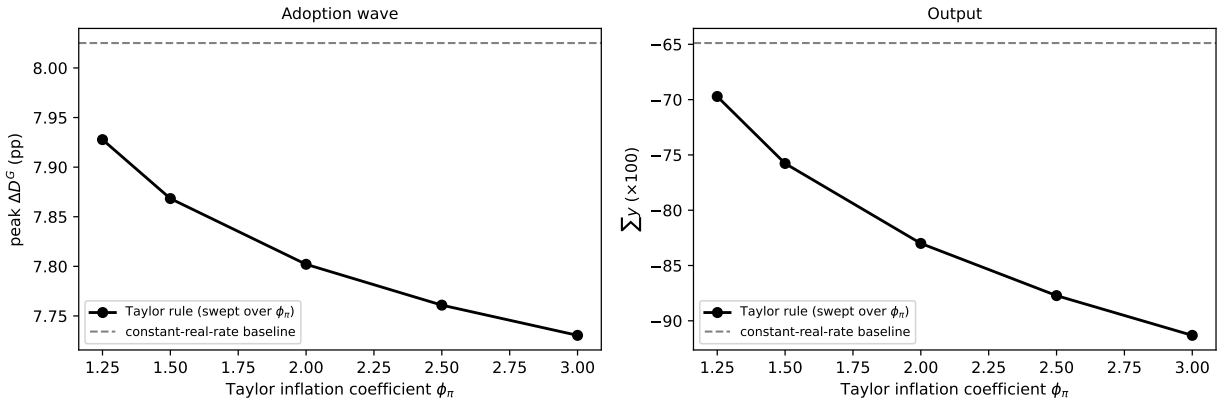
C.2 The monetary stance and the adoption wave

The reverse question is Dietrich's. In their representative-agent economy green durables are interest-rate sensitive, so a strict inflation target slows the transition. The mechanism survives here and acquires a distributional dimension: the switch is a lumpy purchase financed against the borrowing constraint, so its user cost rises with the real rate, and it does so most for the lower-wealth households that carry the crisis-induced adoption margin (Figure 2).

The stance operates on the wave, but weakly. Holding the brown-price shock fixed and varying the rule, a Taylor rule that leans against the energy-driven inflation ($\phi_\pi = 1.5$) raises the ex-ante real rate, by +4.5 cumulated over $H = 24$, and shrinks the peak green-share response from 8.03 to 7.87 pp. A deliberate 25-basis-point accommodation does the opposite, lifting it to 8.08 pp and cutting the output loss; the standalone rate cut raises adoption on its own (D^G peak +0.09 pp, consumption +5.8 cumulated) as easier financing relaxes the constraint on the switch. The sign is Dietrich’s; the magnitude is small.

Figure 25 sweeps the inflation coefficient. Stricter targeting shrinks the wave monotonically, from 7.93 pp at $\phi_\pi = 1.25$ to 7.73 at $\phi_\pi = 3$, but the whole range spans a fifth of a percentage point, while the cumulative output loss grows from -70 to -91 . Strict inflation targeting during an energy crisis buys a marginal slowdown of the transition at a first-order output cost.

Figure 25: The monetary stance and the adoption wave

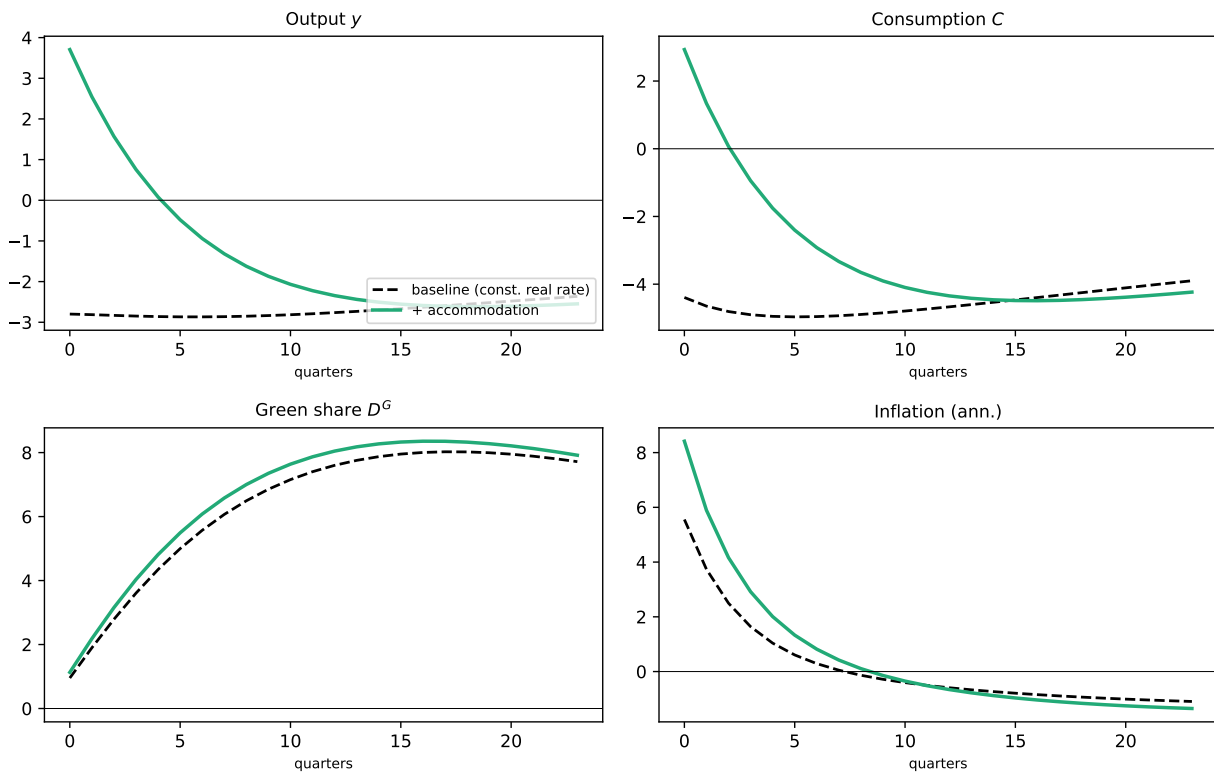


Notes: Brown-price shock, as the Taylor inflation coefficient ϕ_π rises (with $\phi_{\pi e} = 0$). Left: peak ΔD^G . Right: cumulative output. Solid: Taylor rule swept over ϕ_π ; dashed: constant-real-rate baseline.

The comparison that matters is with the fiscal instruments. The peak wave’s semi-elasticity to the real rate is about -0.035 pp per unit of cumulative rate, so moving the wave by one percentage point would take a cumulative rate change of order thirty, an implausible monetary action, whereas the retail price cap removes almost the entire 8.0 pp wave directly (Section 6). Monetary policy reaches the adoption margin only through the financing cost of the switch, a second-order channel; the cap and the transfer reach it through the relative price itself, which is first-order. This is why we keep monetary policy at the constant-real-rate baseline in the main text: not because the stance is neutral for adoption, but because it is dominated, by an order of magnitude, by the instruments the paper ranks.

An accommodative offset. A natural crisis exercise runs an accommodative innovation *alongside* the energy shock, leaning against it rather than waiting. Figure 26 sizes the accommodation to halve the cumulative output loss, which takes a large front-loaded cut (about 144 basis points annualised). It softens the fall in cumulative output (-65 to -32) and consumption (-109 to -76) and lifts the adoption wave (8.0 to 8.4 pp), the financing channel adding to the effect, but pays for the relief with more inflation (impact annualised inflation $+5.6\%$ to $+8.4\%$) and an output path that overshoots on impact before settling. The offset buys output stabilisation and a small transition dividend at the cost of the price level and a much larger rate move than the fiscal instruments require; it does not substitute for the relative-price instruments, which stabilise output without that inflation cost.

Figure 26: Accommodative monetary offset



Notes: Run alongside the brown-price shock, sized to halve the cumulative output loss: energy shock alone (dashed) against energy shock plus accommodation (solid). Level deviations $\times 100$.

D The logit taste scale: smoothing device and structural elasticity

The extreme-value taste shocks on the adoption choice play two roles at once, and this appendix separates them. The first role is computational: the logit smooths the discrete policy, so that choice probabilities and the value function are differentiable with closed-form derivatives, which is what the sequence-space method of [Auclert et al. \(2021\)](#) requires to build Jacobians through the household block ([Iskhakov et al., 2017](#)). In that tradition the scale σ_ε is a numerical device: one takes it small and verifies that results are insensitive to it. The second role is economic. Differentiating the logit gives $dP_i = P_i(dV_i - dV)/\sigma_\varepsilon$: the response of the choice probabilities to *any* perturbation of the value gap scales with $1/\sigma_\varepsilon$, so the same parameter that smooths the policy is the structural elasticity of the margin. Which of the two readings governs depends on where the population sits on the logit.

If choice probabilities are interior, of order one, as for labour-force participation, where employment rates and quarterly flows are tens of percent, then $P(1 - P)$ is bounded, the $\sigma_\varepsilon \rightarrow 0$ limit is a well-defined deterministic cutoff, and aggregate responses converge to the density of households at the threshold: an object of the wealth and productivity distribution rather than of σ_ε . There the numerical reading is legitimate, the economics of the margin is carried by the distribution (and, in search applications, by the frictions standing between the choice and the outcome), and the smoothing convention applies. If instead the margin is *rare*, choice probabilities deep in the left tail, the logit is exponential there, $d \log P = d\Delta/\sigma_\varepsilon$ holds for every household regardless of its level, and the semi-elasticity of the flow is identically $1/\sigma_\varepsilon$. No feature of the distribution can attenuate it; there is no σ_ε -insensitive range, and the scale *is* the elasticity of the margin.

Our margin is rare. The calibration targets a 5% green share with a five-year durable life, which fixes the steady-state switching flow at $\delta_g \bar{D}^G / (1 - \bar{D}^G) = 0.26\%$ per quarter. [Table 12](#) reports the distribution-weighted quantiles of the quarterly switch probability among brown incumbents: the mean is 0.26%, the median is essentially zero, the 95th percentile is 1.6%, and even the 99.9th percentile stays below 9%. Virtually no mass approaches interior probabilities. The adjustment is lumpy and quasi-irreversible, and the structural discrete-choice literature estimates the scale for such margins from choice data ([Rust, 1987](#)).

Table 12: Switch probabilities among brown incumbents

type	β	mass%	mean	p50	p95	p99.9	max
impatient	0.924	96.9	0.17	0.00	0.77	10.60	37.8
middle	0.954	95.5	0.25	0.00	1.28	12.36	35.3
patient	0.984	93.4	0.37	0.01	2.12	11.81	28.8
all	–	95.2	0.26	0.00	1.22	11.71	37.8

Notes: Distribution-weighted, baseline steady state. Mean, median and upper quantiles of $P(\text{switch})$ in %, by discount-factor type and pooled. The mean equals the steady-state flow identity $\delta_g \bar{D}^G / (1 - \bar{D}^G) = 0.26\%$ (Appendix D).

Three consequences follow, each handled in the paper. First, the adoption-channel magnitudes are proportional to $1/\sigma_\varepsilon$, so a level moment cannot discipline them; Section 4.1 characterises the $(\sigma_\varepsilon, \psi_g)$ locus and identifies the adoption elasticity as the moment that pins the scale. Second, the same $1/\sigma_\varepsilon$ governs the curvature of the margin, so the linearisation error is concentrated in D^G and grows as σ_ε falls; Appendix B.1 quantifies it. Third, the steady state and the policy ordering are invariant along the locus, so none of the paper’s qualitative results depend on the scale. The scale does govern the size of the adoption wave and of the adoption channel, and that size is what the elasticity moment disciplines.

D.1 From micro estimates to the taste scale: the procedure

This subsection documents the calibration of σ_ε step by step, so that every number in Section 4.1 can be audited and reproduced. The entire procedure is implemented in a single runner (`runners/run_taste_identification.py`; `python main.py taste_id` regenerates Table 13, Table 12 and Figure 27); the grid, the perturbation sizes and the dollar mapping are constants at the top of the file.

1. *Trace the locus.* For each σ_ε on the grid $\{0.02, 0.035, 0.05, 0.07, 0.10, 0.15, 0.20\}$, resolve the baseline steady state with ψ_g as the unknown targeting $\bar{D}^G = 5\%$. This delivers $\psi_g^*(\sigma_\varepsilon)$, rising from 0.58 to 1.74, with the steady state, shares, flows, prices, distribution, invariant along the locus.
2. *Switch to the fixed- ψ configuration.* All comparative statics below hold ψ_g fixed at ψ_g^* and let the green share D^G adjust endogenously (the same configuration that builds the ex-ante ETS economy of Section 7.2). Elasticities are computed as finite differences on this configuration around the calibrated point.
3. *Operating-cost margin.* Perturb the delivered green/brown price ratio $\varrho = 0.54$ down by 0.01 at fixed ψ_g^* and measure $\Delta \ln D^G / \Delta \ln(1 - \varrho)$, the elasticity of the green share

to the operating-cost gap. It falls from 4.4 to 0.8 across the grid (2.73 at the baseline).

4. *Upfront-cost margin: the matched moment.* Perturb ψ_g down by 1% at fixed calibration and measure the response of the switching flow $P = D^{sw}/(1 - D^G)$. Per 1% of ψ_g the semi-elasticity is nearly flat in σ_ε (5.9% to 4.2%), because ψ_g^* rises along the locus and compensates; the identifying variation comes from converting to absolute dollars. With one model unit equal to quarterly household consumption $\bar{C}_\$,$ the semi-elasticity per \$1,000 is the per-1% figure times $1,000/(0.01 \psi_g^* \bar{C}_\$)$; at $\bar{C}_\$ = \$20,000$ it falls from 64% to 12% across the grid (Table 13).

The empirical anchors are converted to the same units. [Gallagher and Muehlegger \(2011\)](#) and [Chandra et al. \(2010\)](#) estimate that a \$1,000 tax incentive raises hybrid adoption by 31–38%; no conversion is needed. [Muehlegger and Rapson \(2022\)](#) report a demand elasticity of -2.1 at a mean transaction price of \$26,000, which converts to $2.1 \times 1,000/26,000 \approx 8\%$ per \$1,000; their estimated pass-through of 73–85% implies the price faced by the buyer moves nearly one-for-one with the subsidy, so the conversion needs no incidence correction. The choice between the two anchors is a population statement rather than a taste. At a 5% green share the model’s marginal adopters consume roughly twice the average brown household: they are the higher-income early adopters of the hybrid-era estimates, not the mass-market population of [Muehlegger and Rapson \(2022\)](#), whose estimate therefore serves as the floor on responsiveness rather than the target.

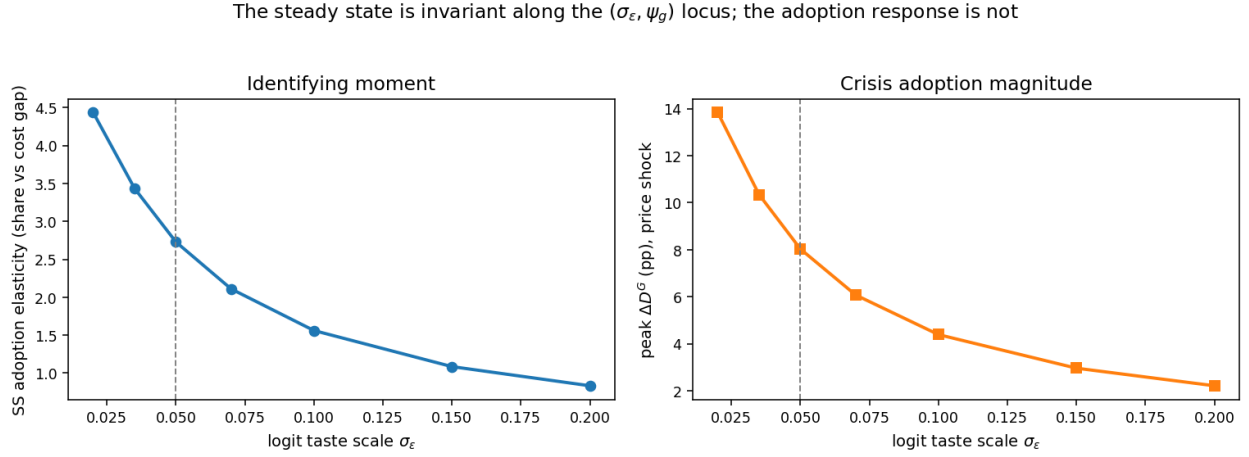
Matching the early-adopter range pins $\sigma_\varepsilon \in [0.05, 0.07]$. The implied premium $\psi_g^* \bar{C}_\$_$ provides an independent cross-check: it is \$15,000–\$18,000 on the pinned interval, within the heat-pump/EV premium range, while $\sigma_\varepsilon \geq 0.10$ would imply premia of \$22,000 and above.

The one free constant is the dollar mapping $\bar{C}_\$_$, an assumption stated rather than estimated. A higher $\bar{C}_\$_$ scales down the model’s per-\$1,000 response, which moves the elasticity-based estimate toward lower σ_ε ; it also scales up the implied premium, which moves the premium-based estimate the same way. The two criteria move together and the joint interval stays narrow. Concretely, at $\bar{C}_\$ = \$16,000$ the joint pin is roughly $[0.07, 0.10]$; at \$20,000, $[0.05, 0.07]$; at \$24,000, $[0.04, 0.05]$. Every one of these intervals lies inside the sensitivity grid of Table 13, over which the steady state and the policy ordering are invariant; what the mapping moves is the size of the adoption channel, bounded on the grid by peak responses of 2 to 14 pp.

The procedure has two limits. The anchors come from vehicle adoption in the United States in the 2000s–2010s; their external validity for heat pumps or for European households is an assumption, and estimating the adoption elasticity directly, rather than importing it,

is the natural next step (Section 9). The anchors also identify the upfront-cost margin; the operating-cost margin, the channel documented by [Beresteanu and Li \(2011\)](#) for gasoline prices and hybrid demand, is disciplined here only through the model’s internal consistency, which ties the two margins together at the calibrated point.

Figure 27: Identification of the taste scale



Notes: Steady state held at $\bar{D}^G = 5\%$ by recalibrating ψ_g ; the dashed line marks the baseline $\sigma_\varepsilon = 0.05$. Left: the steady-state adoption elasticity (green share against the operating-cost gap) against σ_ε . Right: the crisis adoption magnitude (peak ΔD^G to the brown-price shock).

Table 13: Identification of the logit taste scale

σ_ε	ψ_g	premium \$000	cost-gap elast.	%/\$1,000	peak ΔD^G	$\sum y$
0.020	0.577	11.5	4.44	64	13.84	-63.5
0.035	0.671	13.4	3.43	50	10.34	-64.5
0.050	0.770	15.4	2.73	39	8.03	-64.9
0.070	0.905	18.1	2.11	30	6.07	-64.9
0.100	1.105	22.1	1.56	23	4.39	-64.6
0.150	1.427	28.5	1.08	16	2.96	-63.9
0.200	1.736	34.7	0.83	12	2.21	-63.1

Notes: For each σ_ε , ψ_g is recalibrated to hold $D_{ss}^G = 5\%$, leaving the steady state invariant. Columns: the implied green premium in dollars (one model unit = quarterly household consumption of \$20,000); the steady-state elasticity of the green share to the operating-cost gap at fixed ψ_g ; the semi-elasticity of the switching flow to a \$1,000 cut in ψ_g (matched to [Gallagher and Muehlegger, 2011](#); [Chandra et al., 2010](#); [Muehlegger and Rapson, 2022](#)); the peak crisis adoption response; and the cumulative output loss.

E Calibrating the operating-cost ratio ϱ

The single price parameter of the adoption block, $\varrho = \bar{P}_G^E / \bar{P}_B^E$, is a delivered operating cost: the cost of one unit of energy service (a kilometre) from the green durable relative to the brown one. It is not a price per unit of energy (per kilowatt-hour, electricity is dearer than gasoline) but a cost per unit of service, net of each technology’s conversion efficiency. The upfront cost of switching is a separate object, ψ_g .

We calibrate ϱ on the electric-vehicle margin, the durable for which both the operating-cost gap and the adoption elasticity (Section 4.1) are measured. With P^G, P^B the retail electricity and gasoline prices and e_G, e_B real-world energy use (Table 14),

$$\varrho = \frac{P^G e_G}{P^B e_B} = \frac{0.290 \times 0.21}{1.62 \times 0.070} = \frac{0.0609}{0.1134} = 0.54.$$

Table 14: Inputs to ϱ

		Value	Source
Electricity price, households	P^G	0.290 EUR/kWh	Eurostat <code>nrg_pc_204</code> , 2025S2, incl. taxes
Gasoline price, Euro-95	P^B	1.62 EUR/L	EC Weekly Oil Bulletin, 2025S2, incl. taxes
Electric-car energy use	e_G	0.21 kWh/km	IEA <i>Global EV Outlook 2026</i>
Gasoline-car fuel use	e_B	7.0 L/100km	National on-road data (e.g. FR SDES)

Notes: European, tax-inclusive, same semester.

Three choices need stating. (i) *Tax-inclusive prices.* The baseline has no carbon tax, so the household compares retail prices; the gap between the two ratios (0.54 inclusive against 0.88 excluding taxes) is the asymmetric taxation of the two fuels, which the ETS acts on. (ii) *Real-world energy use.* Both figures are on-road or metered rather than type-approval, so charging losses are already in the electricity figure and the two sides are on the same basis. (iii) *Fleet averages.* A matched compact pair (electric 17 kWh/100km, gasoline 6.2 L/100km) gives 0.49, so the ratio does not depend on vehicle-size composition. As an internal check, 0.21 kWh/km is 2.3 litres of gasoline equivalent per 100 km, 68% below the gasoline car’s 7.0, matching the roughly 70% energy saving the same source reports for a comparable electric car. The heating margin (an air-source heat pump against a condensing gas boiler at Eurostat household electricity and gas prices) gives a nearby $\varrho \approx 0.62$, so reading the durable as a heat-pump/EV composite would move the ratio only inside the range below.

Sensitivity. Holding the steady-state green share at $\bar{D}^G = 5\%$, ψ_g re-solves as ϱ varies (Table 15). Across the whole plausible range the policy CEV ranking is unchanged (flat transfer $\succ$ Slutsky $\succ$ ex-post cap $\succ$ ex-ante ETS $\succ$ laissez-faire $\succ$ green subsidy); only the crisis adoption magnitude responds, by about one percentage point of peak green share.

Table 15: Sensitivity of the baseline to ϱ

ϱ	0.48	0.54	0.62	0.70	0.80
ψ_g (solved)	0.827	0.770	0.696	0.625	0.538
peak ΔD^G (pp)	7.74	8.03	8.39	8.74	9.14
peak output loss $-y$ (%)	2.86	2.87	2.88	2.89	2.90

Notes: Price shock, import booking. $\bar{D}^G = 5\%$ imposed throughout, ψ_g re-solved as ϱ varies.